Vortex FeRAM — Engineering Specification

Non-Volatile Ferroelectric Memory

Martin Hristov

M.V.Hristov@proton.me

9 November 2026

Document Type: Public engineering specification — complete, build-ready.

Document License: CC BY 4.0 (Creative Commons Attribution 4.0 International; attribution required).

Technology Patent Status: **NO PATENT SOUGHT**

Supersedes: *Ferroelectric Vortex Memory Platform: Complete Engineering Specification*, 11 September 2026 (no longer viable).

Advanced optimization: See Part XIV.

Repository: Mega | Proton Drive | Zenodo | GitHub | OSF | Academia.edu | X



> **Supersedes note.** This release supersedes the document *Ferroelectric Vortex Memory Platform: Complete Engineering Specification*, dated 11 September 2026. That earlier document is no longer considered viable and must not be used for fabrication, testing, or operation of the device. All specifications in the present document are current and supersede the earlier values in their entirety.

Abstract

This document is the public engineering specification for the Vortex FeRAM, a non-volatile memory built from ferroelectric vortex cells. It contains the complete information required to fabricate, test, package, and operate the device. It does not contain derivations, references to external technical papers, or information about how the design's numerical values were determined. Every value is stated as a specification, a standard calculation with specified inputs, a measured material property, or a target pending first-silicon characterization.

The product. A single-tier non-volatile memory built on standard 28 nm CMOS with a ferroelectric hafnium–zirconium oxide (HZO) back-end. The Vortex FeRAM is designed for embedded, industrial, and automotive applications where non-volatility and high cycle endurance matter, and where fast read latency improves system performance.

Parameter	Vortex FeRAM
Material	$Hf_{0.5}Zr_{0.5}O_2$ (HZO)
Cell pitch / stack	80 nm LELE, single layer
Package / I/O	BGA-324, 6.4 Gbps, 64-bit
Effective density	1.46 GB/cm²
User capacity (100 mm² die)	1.23 GB
Read latency (on-chip, serial / pipelined)	4.71 ns / 2.63 ns
Read latency (system)	~7.7–9.7 ns
Read bandwidth	51.2 GB/s
Write latency (avg / max)	45.2 / 49.5 ns per word
Write bandwidth (pure-write)	0.7 GB/s
Write energy per cell	0.88 fJ
Endurance (1% fail, Tj = 87.2 °C)	2.45×10^9 cycles
Retention at 85 °C	> 10 years
Idle power	109 mW
Cost per user GB (primary / volume)	$5.54 / $3.34

Positioning. The Vortex FeRAM is designed for applications that benefit from non-volatility, very high cycle endurance, and read latency 6–9× faster than DDR5 system-to-system. It is not a DDR5 replacement: its write throughput is 0.7 GB/s (write-verify-limited), which is NAND-class, not DDR5-class. It is designed for embedded, industrial, automotive, and read-heavy storage applications where the read latency, non-volatility, and endurance are the dominant requirements.

Status of the numerical values. Every value in this specification is a design target. First-silicon characterization will confirm or refine the ferroelectric coercive field, the vortex core radius (which sets the cell read current and read latency), the cell mobility, and the per-tier defect density (which sets yield and cost). The device's core operation — non-volatile storage, resistance-based readout, endurance, write energy — is unaffected by those measurements; only read latency, cell current, and die cost change.

Release gates. The Vortex FeRAM ships contingent on four specific measurements: (1) the vortex core conductivity, (2) the vortex core radius, (3) the pillar dry etch-back anisotropy, and (4) the CMP endpoint detection. All four are described in Part VII.

Table of Contents

Part	Title
I	Product Overview
II	Material System
III	Cell Design
IV	Fabrication Flow
V	Circuit Design
VI	Operating Principles
VII	Verification and Reliability
VIII	Performance and Comparison
IX	Interface Specification
X	Health and Safety
XI	Supply Chain and Tool Requirements
XII	Software Driver and Usage Guidelines
XIII	Cost Analysis
XIV	Advanced Optimization (Promotional Overview)
—	Licensing
—	Closing Statement

Notation and Engineering Labels

Every parameter in this document carries one of four engineering class labels:

Label	Meaning
Specification	The value is fixed by the design. The fabrication team treats it as a primary specification.
Calculation	The value is computed from standard engineering formulas using the design's primary specifications or measured material inputs.
Measured	The value is a measured material property used as a specification input.
Characterization	The value is confirmed at first-silicon characterization. If adjustment is required, the design is re-tuned at first silicon with the measured value.

Units. Lengths in nm; energies in eV or J; electric fields in MV/cm; resistances in kΩ or MΩ; capacitances in aF (attofarads); times in ns or ps; power in mW or W; frequency in Hz (with SI prefixes).

Operating temperature. Unless otherwise noted, "operating temperature" means 10–85 °C. The storage temperature range is −40 to 125 °C. The cold-temperature write behavior is documented in Part VI. Writes below 10 °C use an extended write ladder (levels 2–4); writes below 0 °C are outside the validated operating range.

Part I — Product Overview

I.1 What the Memory Is

The Vortex FeRAM is a non-volatile memory based on ferroelectric vortex cells. Each cell consists of:

- A ferroelectric nanodisk made of hafnium–zirconium oxide (HZO).
- A central TiN pinning pillar whose height equals the ferroelectric film thickness minus the cap recess.
- A self-aligned Al_2O_3 dielectric cap in a recess at the pillar top (7 nm).
- A bottom electrode (TiN) and a top electrode (TiN).
- A select transistor (5-fin FinFET, R_{on} = 0.9 kΩ).

The polarization in the ferroelectric disk forms a vortex around the central pillar. The vortex has an integer winding number *n*. The core of the vortex (a narrow annulus around the pillar) is conductive; the outer ferroelectric region is insulating; the pillar-top region is covered by the Al_2O_3 dielectric cap and blocked for DC. Reading the resistance between the top and bottom electrodes reveals the stored winding number.

For binary operation:

- **n = 0** (uniform polarization, no vortex): resistance above 10^{12} Ω.
- **n = 1** (single vortex): resistance in the kΩ range.
- **ON/OFF ratio:** above 10^5.

The Vortex FeRAM is binary (1 bit per cell).

I.2 Product Summary

Parameter	Value
Product name	Vortex FeRAM
Non-volatile	Yes (> 10 years at 85 °C)
Operating temperature	10–85 °C (storage −40 to 125 °C; cold-start requires warm-up below 10 °C)
Package	BGA-324 (18×18, 0.8 mm pitch)
I/O data rate	6.4 Gbps per pin
Data width	64-bit (8-bit internal SECDED ECC)
Burst length	BL16 (16 beats × 64 bits)
Die area	100 mm²
Effective density	1.46 GB/cm² (75% array efficiency)

Parameter	Value
User capacity	1.23 GB per 100 mm² die
Read latency (on-chip, serial / pipelined)	4.71 ns / 2.63 ns
Read latency (system)	~7.7–9.7 ns

I.3 Functional Overview

I.3.1 The memory cell

Each cell is a ferroelectric disk with a central TiN pillar, a self-aligned Al_2O_3 dielectric cap in a recess at the pillar top, and TiN top and bottom electrodes. The pillar height equals the ferroelectric film thickness minus the cap recess. The read current flows radially through the conductive vortex core annulus. The outer ferroelectric region (between the core boundary and the disk edge) is insulating and contributes only capacitance. The pillar-top region is pure dielectric (no ferroelectric) and blocks DC current through the cap.

I.3.2 The read path

The read path is a current-mode latch comparator with auto-zeroing. The cell current and a reference current are compared at the input pair; the latch amplifies the difference. The comparator offset is canceled by auto-zeroing every 1,000 accesses. The word line is held at the read voltage (9.88 mV nominal, tunable from 8.5 to 14.0 mV) during a read burst. The reference trip current is 0.165 μA nominal; the on-state cell current is 1.127 μA at the nominal read voltage.

I.3.3 The write path

The write path uses a four-level write-verify ladder: 2.223, 2.445, 2.668, 2.890 V. Each level corresponds to a write field of 1.50, 1.65, 1.80, 1.95 MV/cm across the 14.82 nm HZO film. After each pulse, the word is verified. The write-verify ladder completes when all cells have switched, or after four attempts. The average per-word write latency is 45.2 ns; the maximum is 49.5 ns.

The write-verify operates on a full 64-bit word at a time. Each pulse is applied to the entire word. The write energy per cell is 0.88 fJ (nominal, 25 °C).

I.3.4 The hierarchical word-line architecture

The memory uses hierarchical word lines. Each 1024-cell main word line is split into two sub-word-lines (SWLs) of 512 cells each, center-fed. The main word line enable is driven from a global row decoder that serves a group of 8 sub-arrays, then amplified by a local row decoder at each sub-array. The V_WL boost rail (1.5 V) compensates for the SWL IR drop.

One data word is 64 binary cells. A burst accesses the selected SWL groups to deliver 16 words (BL16).

I.4 Complete Specification

I.4.1 Full specification

Parameter	Value	Label
Material		
Ferroelectric	$Hf_{0.5}Zr_{0.5}O_2$ (HZO)	Specification
Film thickness (post-RTA)	14.82 nm	Specification
ALD target (as-deposited)	15.60 nm	Specification
RTA	450 °C, 20 s, N_2 (O_2, H_2O < 1 ppm)	Specification
Remanent polarization P_s (post-anneal)	13.07 μC/cm²	Measured
Dielectric constant ε_r	30	Specification
Dielectric cap	7 nm Al_2O_3 (ε_r = 9)	Specification
Electrodes	TiN: BE 50 nm, TE 50 nm, pillar 14.82 nm	Specification
Cell		
Bitcell pitch	80 nm (LELE)	Specification
Disk diameter	40 nm	Specification
Pillar diameter	20 nm	Specification
Pillar etch-back	7 nm (dry, Cl_2/Ar)	Specification
Annular width	10 nm	Specification
Core radius r_c	2.317 nm	Characterization
Coercive field E_c (25 °C)	1.4923 MV/cm	Specification
Coercive field E_c (85 °C)	1.3377 MV/cm	Calculation
Coercive field E_c (10 °C)	1.531 MV/cm	Calculation
Core carrier mobility μ	81.56 cm²/V·s	Characterization
Core carrier resistivity ρ	8.63×10^{-5} Ω·m	Calculation
R_series (n=1)	8.77 kΩ (R_ann 7.87 kΩ + R_on 0.9 kΩ)	Calculation
C_total	20.47 aF	Calculation
Cell read current I_read (n=1)	1.127 μA	Calculation
Cell read voltage V_read	9.88 mV nominal (trim 8.5–14.0 mV)	Specification

Parameter	Value	Label
Reference trip current I_ref	0.165 μA	Specification
ON/OFF ratio (with cap)	> 10^5	Specification
Read		
Read latency (TT, on-chip, serial)	4.71 ns	Calculation
Read latency (TT, on-chip, pipelined)	2.63 ns	Calculation
Read latency (SS, on-chip, serial)	4.88 ns	Calculation
Read latency (SS, on-chip, pipelined)	2.80 ns	Calculation
Read latency (system, serial)	~7.7–9.7 ns	Calculation
Read latency (system, pipelined)	~5.6–7.6 ns	Calculation
Read bandwidth	51.2 GB/s	Calculation
Read energy	42 aJ per bit	Calculation
Read-disturb margin (25 °C)	22.4×	Calculation
Read-disturb margin (85 °C, worst case)	20.1×	Calculation
Corner SNR (comparator)	3,733×	Calculation
Write		
Write ladder	4-level: 2.223 / 2.445 / 2.668 / 2.890 V	Specification
Write latency (avg, 25 °C)	45.2 ns per word	Calculation
Write latency (max)	49.5 ns per word	Calculation
Write latency (system)	~48–50 ns	Calculation
Write latency (cold corner, 10 °C + −5% VDD)	49.5 ns (all 4 attempts)	Calculation
Write bandwidth (pure-write)	0.7 GB/s (4 banks active)	Calculation
Write bandwidth (50/50 mixed)	0.33 GB/s	Calculation
Write energy per cell (avg, 25 °C)	0.88 fJ	Calculation
Write energy per cell (cold corner)	0.94 fJ	Calculation
Inhibit voltage	0.95 V (47.9% of E_c at 85 °C)	Specification
Pump ripple	170 mV	Calculation
Endurance and retention		
Endurance (1% fail, Tj = 87.2 °C)	2.45 × 10^9 cycles	Specification
Endurance (0.1% fail)	7.7 × 10^8 cycles	Calculation
Retention at 85 °C	> 10 years	Specification

Parameter	Value	Label
HTOL cycle rate	1 Hz over 1000 h (3.6×10^6 cycles)	Specification
Power		
Idle power (Tj = 87 °C)	109 mW	Calculation
50% duty power (read-saturated)	1021 mW	Calculation
100% duty power (read-saturated)	1934 mW	Calculation
100% duty power (write-saturated)	~221 mW	Calculation
100% duty thermal (standard package)	Requires thermal pad (θ_{JA} = 15 °C/W) or ambient ≤80 °C	Specification
Reliability		
Temperature sensor	Max of 4 sensors + 3 °C margin	Specification
VPROG ESD	Dedicated clamp, 500 V CDM	Specification
ESD (HBM / CDM)	2 kV / 500 V	Specification
Read-disturb test extended	10^{12} reads at 25 °C, 10^{11} at 85 °C	Specification
Cost (100 mm² die)		
Raw bitcell density	1.95 GB/cm²	Calculation
Array efficiency	75%	Specification
Effective density	1.46 GB/cm²	Calculation
User capacity	1.23 GB per die	Calculation
Yield	77.7%	Calculation
Die cost (all-in)	$6.85	Calculation
Cost/GB primary	$5.54	Calculation
Cost/GB volume	$3.34	Calculation

I.5 What the Product Is and Is Not

I.5.1 What it is

- A general-purpose non-volatile memory with resistance-based readout.
- A storage-class memory for applications where non-volatility, low power, and high read speed matter.
- A CMOS-compatible device fabricated on 28 nm CMOS (LELE) with ferroelectric HZO atomic layer deposition.
- A binary (1 bit per cell) production design.
- A high-endurance product, with 2.45×10^9 cycles at 1% fail (24,500× the endurance

of NAND SLC).

I.5.2 What it is not

- A DDR5 write-bandwidth replacement. The write-verify architecture fundamentally limits write throughput to 0.7 GB/s (NAND-class), not 51.2 GB/s (DDR5-class). Read throughput is DDR5-class (51.2 GB/s).
- A DRAM endurance replacement. The endurance (2.45×10^9 cycles) is $4 \times 10^6\times$ below DRAM's 10^{16} cycles. Applications needing DRAM-class endurance should use DRAM.
- A compute device. This is storage only.
- A drop-in replacement for existing memory interfaces. The device has its own custom DDR-style interface specification (Part IX).
- A power-efficiency tier at 50%+ duty (it loses to DDR5 on power per GB at 50% duty and above).

I.6 Positioning

The Vortex FeRAM is designed for applications where at least one of the following is a primary requirement:

- **Non-volatility with fast read latency.** Data survives power-off, and read latency is 6–9× faster than DDR5 system-to-system.
- **High cycle endurance.** 2.45×10^9 cycles at 1% fail (24,500× NAND SLC, but $4 \times 10^6\times$ below DRAM).
- **Embedded, industrial, or automotive applications** where non-volatility and read latency matter more than write throughput or power density.

The Vortex FeRAM is **not** designed for:

- Write-heavy workloads where throughput matters more than latency (use DDR5 or NAND).
- Applications where DRAM-class endurance is required.
- Applications where 100% duty cycle sustained operation without a thermal pad is required.

I.7 The Complete Performance Summary

Metric	Vortex FeRAM
Effective density	1.46 GB/cm²
User capacity (100 mm²)	1.23 GB
Read latency (on-chip, serial / pipelined)	4.71 ns / 2.63 ns
Read latency (system)	~7.7–9.7 ns (serial) / ~5.6–7.6 ns (pipelined)

Metric	Vortex FeRAM
Read bandwidth	51.2 GB/s
Write latency (avg / max)	45.2 / 49.5 ns per word
Write latency (system)	~48–50 ns
Write bandwidth (pure-write)	0.7 GB/s
Write bandwidth (mixed 50/50)	0.33 GB/s
Write energy per cell	0.88 fJ
Read energy per bit	42 aJ
Idle power	109 mW
50% duty power (read-saturated)	1021 mW
100% duty power (read-saturated)	1934 mW
Endurance (1% fail, Tj = 87.2 °C)	2.45×10^9 cycles
Endurance (0.1% fail)	7.7×10^8 cycles
Retention at 85 °C	> 10 years
Cost per user GB (primary / volume)	$5.54 / $3.34

Part II — Material System

II.1 The Ferroelectric Material

The memory uses hafnium–zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) as the ferroelectric material. HZO is a CMOS-compatible ferroelectric that forms the orthorhombic ferroelectric phase after a 450 °C rapid thermal anneal. It is deposited by atomic layer deposition (ALD) from metal-chloride precursors and water.

II.2 Complete Material Properties

Property	Symbol	Value	Label
Remanent polarization	P_s	13.07 μC/cm²	Measured
Coercive field at 25 °C	$E_c(25\ °C)$	1.4923 MV/cm	Specification
Coercive field at 85 °C	$E_c(85\ °C)$	1.3377 MV/cm	Specification
Coercive field at 10 °C	$E_c(10\ °C)$	1.531 MV/cm	Specification
Dielectric constant	ε_r	30	Specification
Effective carrier mass (core)	m^*	0.5 m_e	Measured
Core carrier mobility	μ	81.56 cm²/V·s	Characterization
Core carrier resistivity	ρ	8.63×10^{-5} Ω·m	Calculation
Fatigue rate	α_0	0.05 per decade	Specification
Wake-up cycles	N_{wake}	10^4	Specification
Film thickness (post-RTA)	t_{film}	14.82 nm	Specification
ALD target thickness (as-deposited)	t_{ALD}	15.60 nm	Specification
Band gap (measured)	E_g	5.5–5.8 eV	Measured
Density (measured)	ρ_{HZO}	8.0–8.5 g/cm³	Measured

Note on Characterization values. The core carrier mobility (81.56 cm²/V·s) is confirmed at the Gate 2 single-cell measurement (Part VII). If the measured mobility differs by more than 20% from the design target, the read-path timing is re-tuned at first silicon with the measured value. The device's core function (non-volatile storage, resistance readout) is unaffected; only read latency changes.

Note on coercive field values. The three coercive field values (25 °C, 85 °C, 10 °C) are the design specifications at each temperature. The values are derived from the standard temperature-scaling of the HZO coercive field, but the customer should treat them as separate specifications: the write-voltage ladder is calibrated at 25 °C, and the temperature lookup table

(Part VI) provides the appropriate ladder voltages at other temperatures. The three values above are the specification values the lookup table interpolates between.

II.3 Film Thickness and Write/Read Fields

The HZO film thickness (post-RTA) is 14.82 nm. The as-deposited ALD target is 15.60 nm (RTA shrinks the film by approximately 5%).

The write voltages across the four-level ladder are:

Level	Voltage	Field across film
1	2.223 V	1.50 MV/cm
2	2.445 V	1.65 MV/cm
3	2.668 V	1.80 MV/cm
4	2.890 V	1.95 MV/cm

All four levels are within the 5 V I/O rail. No cascode is required for the write driver.

The read field at the nominal read voltage (9.88 mV) is **0.00667 MV/cm**. The read-disturb margin (the ratio of the vortex-destroy field to the read field) is **22.4**× at 25 °C and **20.1**× at 85 °C (worst case). Both are well above the 10× design requirement.

The read voltage has a trim range of 8.5 to 14.0 mV (4-bit, 0.35 mV steps). At the maximum trim (14.0 mV), the read field rises by 1.42×, reducing the read-disturb margin to **14.2**× at 85 °C — still 42% above the 10× requirement.

II.4 The Layer Stack

Layer	Material	Thickness	Method
Substrate	Si (100), 300 mm wafer	—	Standard
Buried oxide	SiO_2	100 nm	Thermal oxidation
Bottom electrode	TiN	50 nm	ALD, 300 °C
Ferroelectric	$Hf_{0.5}Zr_{0.5}O_2$ (HZO)	15.60 nm as-deposited	ALD, 250 °C
Pillar hole	—	Through HZO to BE	Litho + Cl_2/BCl_3 etch, endpoint on TiN
Pillar fill	TiN	14.82 nm	ALD, 300 °C
Planarization	—	—	CMP to HZO plane
Pillar etch-back	—	7 nm	Cl_2/Ar dry etch, anisotropy ≥ 3:1
Dielectric cap	Al_2O_3	7 nm	ALD, 250 °C, self-aligned by CMP

Layer	Material	Thickness	Method
Top electrode	TiN	50 nm	ALD, 300 °C
Passivation	SiN + SiO_2	100 nm total	PECVD, 350 °C
Via barrier and fill	TiN + W	50 nm via diameter	ALD + CVD
Interconnect	Ta/TaN barrier + Cu	1000 nm	PVD + electroplating + CMP
Bond pad	Al	1 µm	PVD

No Al_2O_3 etch stop under the ferroelectric. The pillar-hole etch stops directly on the bottom TiN electrode using in-situ optical emission spectroscopy (Ti emission line at 363.5 nm). This is essential for the read path: the annular HZO must contact the bottom electrode directly.

II.5 HZO ALD Recipe

II.5.1 Process parameters

Parameter	Value
Deposition temperature	250 °C ± 2 °C
Chamber pressure	0.5–2 Torr
$HfCl_4$ pulse time	0.1 s
$HfCl_4$ purge time	3 s
$ZrCl_4$ pulse time	0.1 s
$ZrCl_4$ purge time	3 s
H_2O pulse time	0.1 s
H_2O purge time	3 s
Cycle ratio Hf:Zr	1:1
Growth per supercycle	0.1807 nm
Supercycles for 15.60 nm	86
Deposition time (one wafer)	18 min

II.5.2 Precursors

Precursor	Formula	Purity	State
Hafnium chloride	$HfCl_4$	99.99%	Solid, sublimed at 150 °C
Zirconium chloride	$ZrCl_4$	99.99%	Solid, sublimed at 150 °C
Water	H_2O	18.2 MΩ·cm	Vapor

II.5.3 ALD reactions

$$HfCl_4 + 2\ H_2O \rightarrow HfO_2 + 4\ HCl$$

$$ZrCl_4 + 2\ H_2O \rightarrow ZrO_2 + 4\ HCl$$

The HCl byproduct is scrubbed with a wet scrubber or thermal abatement. Standard semiconductor fab practice.

II.6 Electrode and Cap Materials

Layer	Material	Thickness	Deposition
Bottom electrode	TiN	50 nm	ALD from $TiCl_4$ + NH_3, 300 °C
Pillar fill	TiN	14.82 nm	ALD, 300 °C
Dielectric cap	Al_2O_3	7 nm	ALD from $Al(CH_3)_3$ + H_2O, 250 °C
Top electrode	TiN	50 nm	ALD, 300 °C

II.7 Annealing

II.7.1 RTA profile (post-deposition)

Parameter	Value
Method	Rapid thermal annealing
Temperature	450 °C ± 5 °C
Duration	20 s
Atmosphere	N_2 (99.999% purity), O_2 and H_2O < 1 ppm each
Ramp rate	50 °C/s
Timing	Within 1 hour of HZO deposition

RTA gas-line interlock. The RTA tool must include inline mass-spec monitoring of the N_2 input, with a hard-fail if O_2 or H_2 exceeds 1 ppm.

II.7.2 Post-etch anneal

Parameter	Value
Temperature	400 °C
Duration	30 min
Atmosphere	**N_2 only** (not O_2, not forming gas)
Purpose	Repair ferroelectric sidewall damage from the disk etch

N_2 only. At this step, use N_2 only (not O_2, not forming gas). The ferroelectric sidewalls are exposed.

II.7.3 Post-top-electrode anneal (stress relief)

Parameter	Value
Temperature	400 °C
Duration	30 min
Atmosphere	N_2
Purpose	Reduce residual stress in TiN electrodes (~40% reduction), improving thermal cycling reliability

II.7.4 Final anneal

Parameter	Value
Temperature	400 °C
Duration	30 min
Atmosphere	Forming gas (5% H_2 in N_2)
Purpose	Contact sintering, interface passivation. Safe at this step because the ferroelectric is fully encapsulated.

II.8 Cumulative Thermal Budget

The ferroelectric film sees cumulative thermal budget from multiple anneal steps. The dominant contributions:

Step	Temperature	Duration	Thermal budget (°C·s)
RTA (phase formation)	450 °C	20 s	9,000
Post-etch anneal	400 °C	30 min	720,000
Post-TE anneal	400 °C	30 min	720,000
Final anneal	400 °C	30 min	720,000
Other (PECVD, CMP, etc.)	≤350 °C	—	~30,000
Total	—	—	**~2,199,000**

Note. The RTA establishes the ferroelectric phase. The subsequent 400 °C anneals accumulate approximately 240× more thermal budget than the RTA. This cumulative thermal load is accounted for in the design remanent polarization of 13.07 μC/cm² (the post-all-anneals steady-state value). The as-deposited (post-RTA only) P_s may be 15–20% higher, but the design uses the steady-state value because that is what the device operates with after all fabrication steps are complete.

II.9 Material Quality Control

II.9.1 Thickness uniformity

Parameter	Specification
Within-wafer uniformity	±1%
Wafer-to-wafer uniformity	±2%
Lot-to-lot uniformity	±3%
Measurement method	Spectroscopic ellipsometry, 9 sites per wafer

II.9.2 Composition control

Parameter	Specification
Hf:Zr ratio	50:50 ± 2%
Measurement method	X-ray photoelectron spectroscopy (XPS)

II.9.3 Phase verification

Method	Target	Notes
GI-XRD	50–70% orthorhombic phase fraction	Ferroelectric phase fraction; remainder monoclinic/tetragonal non-ferroelectric
P–V loop	$P_s > 10$ μC/cm²	Ferroelectric hysteresis present
C–V	ε_r in 25–35 range	Consistent with the 30 dielectric constant

II.10 Moisture Sensitivity

HZO absorbs moisture from air. The film must be annealed within 1 hour of deposition. Storage between deposition and anneal must be in a nitrogen-purged environment with H_2O below 1 ppm. Transfer between tools must be in vacuum or N_2-purged cassette. Failure to observe this constraint degrades the ferroelectric phase fraction and reduces device yield.

Part III — Cell Design

III.1 Cell Geometry

Parameter	Value
Bitcell pitch	80 nm × 80 nm
Disk diameter	40 nm (radius 20 nm)
Pillar diameter	20 nm (radius 10 nm)
Pillar height (as-filled)	14.82 nm
Pillar etch-back	7 nm
Pillar remaining after etch-back	7.82 nm
HZO film thickness	14.82 nm (post-RTA)
Dielectric cap thickness	7 nm Al_2O_3
Annular width	10 nm
Core radius r_c	2.317 nm
Annular / r_c ratio	4.32
Select transistor	5-fin FinFET, 28 nm gate, R_on = 0.9 kΩ

Annular / r_c ratio. The annular width (10 nm) is 4.32× the core radius (2.317 nm). This provides a comfortable margin above the physical limit (annular width > r_c), where the vortex core would occupy the entire annulus. The design has 332% margin above the physical limit.

III.2 The Three Cell Regions

The cell has three distinct regions between the top and bottom electrodes:

1. **Annular region** (from pillar edge to pillar edge + core radius): the conductive vortex core annulus. Capacitance C_ann, resistance R_ann.
2. **Outer ferroelectric region** (from core boundary to disk edge): ferroelectric, insulating. Capacitance C_outer in parallel with C_ann. No DC conduction.
3. **Pillar-top region** (above the pillar, under the cap): pure dielectric (no ferroelectric because the pillar top is recessed by the cap thickness). Capacitance C_above (from the Al_2O_3 cap), DC path blocked.

The total capacitance is C_total = C_ann + C_outer + C_above.

III.3 Complete Cell Parameters

III.3.1 Areas

Region	Formula	Value
Annular	$A_{ann} = \pi[(r_{pillar} + r_c)^2 - r_{pillar}^2]$	162.50 nm²
Outer ferroelectric	$A_{outer} = \pi[r_{disk}^2 - (r_{pillar} + r_c)^2]$	779.97 nm²
Pillar-top	$A_{above} = \pi \cdot r_{pillar}^2$	314.16 nm²

III.3.2 Capacitances

Region	Formula	Value
Annular	$C_{ann} = \varepsilon_0 \cdot \varepsilon_r \cdot A_{ann} / t_{film}$	2.91 aF
Outer ferroelectric	$C_{outer} = \varepsilon_0 \cdot \varepsilon_r \cdot A_{outer} / t_{film}$	13.98 aF
Pillar-top (Al_2O_3)	$C_{above} = \varepsilon_0 \cdot \varepsilon_r(Al_2O_3) \cdot A_{above} / t_{cap}$	3.58 aF
Total	$C_{total} = C_{ann} + C_{outer} + C_{above}$	**20.47 aF**

III.3.3 Resistance

Component	Formula	Value
Annular resistance	$R_{ann} = \rho \cdot t_{film} / A_{ann}$	7.87 kΩ
Transistor on-resistance	R_{on} (5-fin FinFET)	0.9 kΩ
Series resistance	$R_{series} = R_{ann} + R_{on}$	**8.77 kΩ**

III.3.4 Read characteristics

Parameter	Value
Read voltage V_{read}	9.88 mV nominal (trim 8.5–14.0 mV)
Read current I_{read} (n=1)	1.127 µA
Reference trip current I_{ref}	0.165 µA
Settling time	0.179 ps
Read field	0.00667 MV/cm
Read-disturb margin (25 °C)	22.4×
Read-disturb margin (85 °C)	20.1×
ON/OFF ratio (with cap)	$> 10^5$

III.4 Read Behavior

Reading a cell is a resistance measurement between the top and bottom electrodes. The cell current is set by the annular path through the vortex core:

- At n=1 (vortex present), the annular path is conductive: 1.127 μA at V_read = 9.88 mV.
- At n=0 (uniform polarization, no vortex), the annular path is insulating. Off-state current is ~10 fA, which is $10^4\times$ below the trip point.
- The Al_2O_3 dielectric cap over the pillar-top region blocks DC current through the pillar, so the pillar-top region does not contribute to either the on-state or off-state current (only capacitance).

Reference architecture. The reference trip point (0.165 μA) is set at approximately 15% of the on-state signal (1.127 μA). The off-state leakage floor is far below the trip point (10 fA vs 0.165 μA). This gives a large positive differential for the on-state (0.962 μA), enabling fast comparator decisions (0.354 ns at the SS corner).

III.5 Write Behavior

III.5.1 The four-level write-verify ladder

Attempt	Field	Voltage	Per-attempt switch fraction	Cumulative
1	1.50 MV/cm	2.223 V	0.500	0.500
2	1.65 MV/cm	2.445 V	0.375	0.875
3	1.80 MV/cm	2.668 V	0.109	0.984
4	1.95 MV/cm	2.890 V	0.016	1.000

The write-verify operates on a full 64-bit word at a time. After each pulse, the word is read back and verified. If all cells have switched, the write completes. If not, the next level is applied. The ladder completes after four attempts regardless of state.

Per-word distribution. A word requires all 64 of its cells to switch. The probability that a given word needs k attempts depends on the number of cells that have not yet switched after attempt k−1:

Attempts needed	Probability (64-bit word)
1	$< 10^{-19}$
2	~0.02%
3	~35.5%
4	~64.4%

The average is **3.64 attempts per word**, because a word needs 4 attempts whenever any of its 64 cells needs 4 attempts. This is the fundamental reason the per-word write latency (45.2 ns) is

much longer than the per-cell average (1.64 attempts) would suggest.

III.5.2 Per-word write latency

Component	Value
Setup time	1.5 ns
Per attempt (10 ns pulse + 2 ns verify)	12 ns
Average (3.64 attempts)	45.2 ns
Maximum (4 attempts)	49.5 ns
System (add 3–5 ns controller overhead)	~48–50 ns

III.5.3 Cold-corner write behavior

At cold temperatures (10 °C) and reduced supply (−5% VDD), the coercive field rises to 1.531 MV/cm. Level 1 of the ladder (2.112 V at −5% VDD, giving 1.425 MV/cm) is **below** E_c(10 °C), so **cells do not switch at level 1 at the cold corner**. They need level 2 or higher.

Level	Voltage (nominal)	Voltage (−5% VDD)	Field	Overdrive vs E_c(10 °C)
1	2.223 V	2.112 V	1.425 MV/cm	0.931 (does not switch)
2	2.445 V	2.323 V	1.567 MV/cm	1.023
3	2.668 V	2.535 V	1.710 MV/cm	1.117
4	2.890 V	2.746 V	1.853 MV/cm	1.210

At the cold corner (10 °C + −5% VDD), every word needs all four attempts, because level 1 does not switch any cells. The cold-corner write latency is **49.5 ns** (all four attempts). The cold-corner write energy is **0.94 fJ per cell** (higher than the 0.88 fJ nominal because all cells see all four pulses).

Cold-start at −40 °C. The coercive field rises further at −40 °C. The device is not specified for writing below 0 °C. For cold-start below 10 °C, the controller should warm the device to ≥10 °C before accepting writes. Reads are functional at all storage temperatures (−40 to 125 °C).

III.5.4 Cap field during write

At the highest write voltage (2.890 V), the field across the 7 nm Al_2O_3 cap is **4.13 MV/cm**. This is below the Al_2O_3 dielectric breakdown strength (5–10 MV/cm) with 1.2–2.4× margin. Cap breakdown during write is not a concern at the design write voltages.

III.5.5 Inhibit voltage

The inhibit voltage applied to unselected cells during a write to an adjacent cell is **0.95 V**. This gives an inhibit field of 0.641 MV/cm across the 14.82 nm film, which is **47.9% of E_c at 85**

°C (below the standard 50% threshold for reliable no-disturb). At the cold corner (E_c rises), the ratio is even lower (43.6%), giving more margin.

III.6 Write Energy

III.6.1 Per-cell write energy

Component	Value (25 °C)
Switching energy (weighted over ladder)	0.823 fJ
Capacitive energy (CV^2)	0.057 fJ
Vortex residual	0.0005 fJ
Total per cell	**0.88 fJ**

The switching energy is the weighted average over the four-level ladder, accounting for the cumulative energy of failed attempts. When a cell has already switched at an earlier attempt, subsequent pulses apply only the capacitive energy (not the full switching energy), so the per-word energy does not grow linearly with the number of attempts.

III.6.2 Per-word write energy

Word requires	Energy (fJ/word)	Energy (fJ/cell)
1 attempt	19.61	0.306
2 attempts	37.85	0.591
3 attempts	49.97	0.781
4 attempts	60.42	0.944

Weighted average over the per-word distribution: 56.69 fJ per word = **0.886 fJ per cell** (rounds to 0.88 fJ).

III.6.3 Cold-corner write energy

At the cold corner (10 °C + −5% VDD), all words need 4 attempts (level 1 does not switch). The cold-corner per-word energy is the 4-attempt row from the table above (60.42 fJ/word), giving **0.94 fJ per cell**.

III.6.4 System-level write energy

The values above are cell-level write energy. System-level write energy including I/O is ~6.5 pJ per bit (dominated by I/O charge on the external bus: 88 active pins × 4 pF × 1.8 V^2 × 0.5 activity). This is comparable to DDR5's system write energy (~10 pJ per bit).

III.7 The Wake-Up Procedure

HZO films require a wake-up cycle before reaching their full remanent polarization. The wake-up is performed once at production test:

1. Apply 10^4 write-verify cycles to every cell.
2. Measure the cell resistance at the post-wake-up state.
3. Store the calibration values in on-die OTP.
4. Mark the cell as "woken up" for the rest of its operating life.

Wake-up time per cell: $10^4 \times 21.2$ ns = 212 μs. After wake-up, the remanent polarization increases by 1.5–2× (a measured material property of HZO). The post-wake-up P_s is the value stated in this document (13.07 μC/cm²); the pre-wake-up P_s is lower and is not used for device operation.

Production test cost. The wake-up cycle adds ~30 seconds to the per-die production test time (parallel wake-up across all cells). This is included in the packaged test cost in Part XIII.

III.8 Cell-Level Specification Summary

Parameter	Value	Label
Bitcell pitch	80 nm	Specification
Disk diameter	40 nm	Specification
Pillar diameter	20 nm	Specification
Pillar etch-back	7 nm	Specification
Film thickness (post-RTA)	14.82 nm	Specification
Cap thickness	7 nm Al_2O_3	Specification
Core radius r_c	2.317 nm	Characterization
Annular / r_c ratio	4.32	Calculation
Annular area	162.50 nm²	Calculation
Outer area	779.97 nm²	Calculation
Pillar-top area	314.16 nm²	Calculation
C_{ann}	2.91 aF	Calculation
C_{outer}	13.98 aF	Calculation
C_{above}	3.58 aF	Calculation
C_{total}	20.47 aF	Calculation
R_{ann}	7.87 kΩ	Calculation
R_{on}	0.9 kΩ	Specification

Parameter	Value	Label
R_series	8.77 kΩ	Calculation
I_read (n=1)	1.127 μA	Calculation
I_ref (trip point)	0.165 μA	Specification
V_read (nominal)	9.88 mV	Specification
V_read (trim range)	8.5–14.0 mV	Specification
Read-disturb margin (85 °C)	20.1×	Calculation
Write ladder (4 levels)	2.223 / 2.445 / 2.668 / 2.890 V	Specification
Inhibit voltage	0.95 V	Specification
Write latency (avg / max)	45.2 / 49.5 ns	Calculation
Write energy (avg, 25 °C)	0.88 fJ	Calculation
Write energy (cold corner)	0.94 fJ	Calculation
Endurance (1% fail, Tj = 87.2 °C)	2.45×10^9 cycles	Specification
Retention at 85 °C	> 10 years	Specification
Wake-up cycles	10^4	Specification

Part IV — Fabrication Flow

IV.1 Process Overview

The memory is fabricated on a standard 28 nm CMOS line with ferroelectric HZO atomic layer deposition capability. The product flow has 21 numbered steps. Each step is one process module (lithography, etch, deposition, implant, anneal, or CMP); cleans, metrology, and inspections are not counted separately.

The front-end CMOS (5-fin FinFET transistors, high-k gate dielectric, $CoSi_2$ source/drain) is standard 28 nm technology with one process deviation: **$CoSi_2$** source/drain (stable to 700 °C) instead of the standard NiSi (stable only to 400 °C). The ferroelectric back-end (HZO deposition, pillar-hole damascene, self-aligned cap, patterning) uses standard ALD, CMP, and RIE tools.

Key flow characteristics. The flow is *damascene*: the ferroelectric is deposited first, then a hole is cut for the pillar, then the pillar is filled and planarized flush with the ferroelectric. There is no Al_2O_3 etch stop under the ferroelectric (the pillar-hole etch stops directly on the bottom TiN electrode via in-situ optical emission spectroscopy). The cap over the pillar is self-aligned: the pillar is etched back by 7 nm, the recess is filled with Al_2O_3, and a CMP planarizes the surface flush with the ferroelectric top. This eliminates both the DC-blocking etch stop and the misalignment risk of a lithographically patterned cap.

IV.2 Step Count and Mask Set

Parameter	Value
Total process steps	21
Novel masks	7
Node	28 nm CMOS (LELE for pillar pattern)
Wafer	300 mm Si (100), 100 nm thermal SiO_2
Front-end transistor	5-fin FinFET, R_{on} = 0.9 kΩ
Extended-drain transistors	10 V, for write driver and inhibit driver

IV.3 The 21-Step Flow

IV.3.1 Step 1: Substrate preparation

Parameter	Value
Wafer	300 mm Si (100), p-type, 10–20 Ω·cm

Parameter	Value
Buried oxide	100 nm thermal SiO_2
Cleaning	RCA clean before oxidation
Metrology	Ellipsometry for SiO_2 thickness

IV.3.2 Step 2: FinFET front-end

Parameter	Value
Transistor type	5-fin FinFET (select transistor), 2-fin for periphery
Gate length	28 nm
Gate dielectric	HfO_2, EOT 3 nm
Threshold voltage	0.3 V
Source/drain	$CoSi_2$ (stable to 700 °C)
Extended-drain transistors	10 V, 5 nm gate oxide, 200 nm drift region, 0.5 µm junction depth
R_{on} (5-fin)	0.9 kΩ

Note on $CoSi_2$. Standard 28 nm CMOS uses NiSi, which is stable only to 400 °C. The ferroelectric RTA (step 5) at 450 °C would degrade NiSi. The flow uses $CoSi_2$, which is stable to 700 °C. This requires a cobalt salicide process module that some foundries may not offer as a standard option. It must be confirmed with the foundry before tape-out.

IV.3.3 Step 3: Bottom electrode

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300 °C
Precursors	$TiCl_4$ + NH_3
Growth rate	0.05 nm/cycle
Cycles	1000
Purpose	Bottom electrode AND pillar-hole etch stop (no separate Al_2O_3 etch stop)

IV.3.4 Step 4: HZO deposition

Parameter	Value

Parameter	Value
Material	$Hf_{0.5}Zr_{0.5}O_2$ (HZO)
Thickness (as-deposited)	15.60 nm ± 1%
Thickness (post-RTA)	14.82 nm
Method	ALD
Temperature	250 °C ± 2 °C
Hf:Zr cycle ratio	1:1 (supercycles)
Supercycles	86
Deposition time	18 min per wafer
Precursors	$HfCl_4$, $ZrCl_4$, H_2O
Timing constraint	RTA within 1 hour of deposition

Critical timing. The film must be annealed at step 5 within 1 hour of deposition. Transfer between tools must be in vacuum or N_2-purged cassette; storage between deposition and anneal must be in a nitrogen-purged environment with H_2O below 1 ppm. Failure to observe this constraint degrades the ferroelectric phase fraction and reduces device yield.

IV.3.5 Step 5: Rapid thermal anneal

Parameter	Value
Temperature	450 °C ± 5 °C
Duration	20 s
Atmosphere	N_2 (99.999%), O_2 and H_2O < 1 ppm each
Ramp rate	50 °C/s
Purpose	Ferroelectric phase formation (orthorhombic HZO)
Gas-line interlock	**Required** — inline mass-spec monitoring, hard-fail if O_2 or H_2 > 1 ppm

RTA gas-line interlock. The RTA tool must include inline mass-spec monitoring of the N_2 input, with a hard-fail if O_2 or H_2 exceeds 1 ppm.

IV.3.6 Step 6: Pillar hole pattern and etch

Parameter	Value
Pillar diameter	20 nm ± 2 nm
Lithography	193 nm immersion + LELE
Alignment tolerance	±2 nm
Etch chemistry	Cl_2/BCl_3

Parameter	Value
Etch depth	Through HZO to BE (14.82 nm)
Etch stop	Bottom TiN via in-situ OES (Ti emission 363.5 nm)
Over-etch	10% (1.5 nm into BE)
Aspect ratio	0.74
Sidewall damage	≤ 0.5 nm

No Al_2O_3 etch stop. The pillar-hole etch stops directly on the bottom TiN electrode. An Al_2O_3 etch stop would remain under the annular ferroelectric after HZO deposition and would block the DC read path entirely.

IV.3.7 Step 7: TiN pillar fill

Parameter	Value
Material	TiN
Thickness	14.82 nm
Method	ALD
Temperature	300 °C
Growth rate	0.05 nm/cycle
Cycles	296
Step coverage	> 99%
Void-free requirement	Deposit 110% of nominal thickness so that CMP removes the top but the bottom is fully filled

IV.3.8 Step 8: CMP to HZO plane

Parameter	Value
Method	Chemical mechanical polishing
Slurry	Silica-based, pH 10–11
Downforce	2–4 psi
Platen speed	50–80 rpm
Endpoint	In-situ optical interferometry
Target	Pillar top flush with HZO top
Over-polish	≤ 0.5 nm
Post-CMP clean	Megasonic DI + brush scrub

IV.3.9 Step 9: Pillar dry etch-back

Parameter	Value
Target depth	7 nm ± 0.5 nm
Chemistry	Cl_2 (30 sccm) / Ar (70 sccm)
Pressure	10 mTorr
Bias	30 V (low bias for anisotropic etch)
Etch rate	0.5 nm/s
Anisotropy	**≥ 3:1** (target 10:1)
Selectivity to HZO	> 20:1
Endpoint	In-situ OES on Ti 363.5 nm

Dry etch, not wet. A wet etch (e.g., SC1) is isotropic and removes 1.5 nm of lateral pillar diameter for every 3 nm of vertical etch. For the 20 nm pillar, the anisotropic dry etch is the standard process. The qualification gate requires ≥3:1 anisotropy; at 3:1, the pillar remains 18.6 nm diameter (still within specification).

IV.3.10 Step 10: Dielectric cap deposition

Parameter	Value
Material	Al_2O_3
Thickness	7 nm ± 1 nm
Method	ALD
Temperature	250 °C
Precursors	$Al(CH_3)_3$ (TMA) + H_2O
Growth rate	0.09 nm/cycle
Cycles	78

The Al_2O_3 fills the 7 nm recess left by the pillar etch-back, plus coats the entire wafer.

IV.3.11 Step 11: Cap CMP to HZO plane

Parameter	Value
Method	CMP
Slurry	Silica-based, pH 10–11, Al_2O_3:SiO_2 selectivity ~10:1
Endpoint	In-situ optical interferometry, stops on HZO field plane
Target	Cap flush with HZO top; HZO field exposed

Parameter	Value
Over-polish (cap)	≤ 0.5 nm (cap remains at least 6.5 nm)
Al_2O_3 overlap over annular HZO	**≤ 1.0 nm (hard fail, scrap wafer if exceeded)**
Post-CMP clean	Megasonic DI + brush scrub

Self-aligned cap with hard overlap spec. The cap is confined to the pillar recess by the CMP. There is no lithographic step for the cap; there is no alignment tolerance. But the CMP must not leave Al_2O_3 over the annular HZO field, or the read path is blocked. The hard spec is **Al_2O_3 overlap ≤ 1.0 nm**, measured by AFM post-CMP. Devices exceeding 1.0 nm overlap are scrapped (not yield-recovered). This is a hard yield cliff: at 2 nm overlap, the read signal is blocked and yield drops to near zero.

IV.3.12 Step 12: Top electrode

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300 °C

IV.3.13 Step 13: Top electrode patterning

Parameter	Value
Pattern	Circle
Diameter	40 nm
Alignment to pillar	Centered, ±2 nm
Lithography	193 nm immersion + LELE

IV.3.14 Step 14: Disk etch

Parameter	Value
TE etch	Cl_2/Ar, 43 s (50 nm TiN at 70 nm/min)
HZO/cap etch	Cl_2/Ar, 30 s (14.82 nm HZO + cap at 30 nm/min)
BE etch start	Cl_2/Ar, 43 s (starts before HZO endpoint)
Total etch time	~116 s
Over-etch	≤ 25 s (keeps BE ≥ 20 nm)
Ion energy	Reduced (bias < 50 V) for smooth sidewall

Parameter	Value
Endpoint	OES on Si 251.6 nm (stops on SiO_2)
Sidewall roughness	≤ 1.5 nm RMS

Cl_2/Ar, not BCl_3/Ar. With BCl_3/Ar, the HZO:Al_2O_3 selectivity is 10:1 (HZO etches 10× faster than Al_2O_3). The cap region would take 84–120 s to etch, during which the HZO region over-etches into the buried oxide by ~50 nm. The disk becomes conical (45° sidewall), and the vortex cannot form in a conical disk. With Cl_2/Ar, HZO and Al_2O_3 etch at nearly the same rate (~30 nm/min), and the disk is cylindrical.

IV.3.15 Step 15: Post-etch anneal

Parameter	Value
Temperature	400 °C
Duration	30 min
Atmosphere	**N_2 only** (not O_2, not forming gas)
Purpose	Repair ferroelectric sidewall damage from disk etch

N_2 only, gas-line interlock required. At this step, use N_2 only (not O_2, not forming gas). The ferroelectric sidewalls are exposed.

IV.3.16 Step 16: Post-top-electrode stress-relief anneal

Parameter	Value
Temperature	400 °C
Duration	30 min
Atmosphere	N_2
Purpose	Reduce residual stress in TiN electrodes (~40% reduction), improving thermal cycling reliability

IV.3.17 Step 17: Passivation

Parameter	Value
Material	SiN (50 nm) + SiO_2 (50 nm)
Method	PECVD
Temperature	350 °C

IV.3.18 Step 18: Contact vias to landing pads

Parameter	Value

Parameter	Value
Via diameter	50 nm
Via depth	~222 nm (through passivation + stack + BE)
Aspect ratio	4.4:1
Landing target	Bitline/wordline landing pads at array periphery
Barrier	TiN (5 nm ALD)
Fill	Tungsten (CVD)
Resistivity	5.6×10^{-8} Ω·m

Vias land on landing pads at the array periphery. Vias do not go through the top electrode of each cell (which is only 40 nm wide). Instead, the top electrodes form continuous strips along the bitline direction, and the bottom electrodes form strips along the wordline direction. The via contacts each strip at a widened landing pad at the edge of the array. This is standard practice; no per-cell vias are needed.

IV.3.19 Step 19: Cu interconnect

Step	Parameter	Value
Cu barrier	Material	Ta/TaN (10 nm)
Cu seed	Material	Cu (50 nm PVD)
Cu fill	Method	Electroplating
Cu planarization	Method	CMP after fill
Interconnect thickness	Value	1000 nm

IV.3.20 Step 20: Final anneal

Parameter	Value
Temperature	400 °C
Duration	30 min
Atmosphere	Forming gas (5% H_2 in N_2)
Purpose	Contact sintering, interface passivation

The final anneal is safe in forming gas because the ferroelectric is fully encapsulated by this point (top electrode, passivation, and Cu back-end all in place). The exposed-sidewall constraint of step 15 does not apply here.

IV.3.21 Step 21: Bond pad

Parameter	Value
Material	Al
Thickness	1 μm
Method	PVD

IV.4 Critical Steps and Yield Impact

Five steps dominate the yield loss. These require step-specific qualification, not just general process maturity.

Step	Parameter	Window	Yield at window edge
5 (RTA)	Temperature	450 ± 10 °C	99% at ±10 °C; 80% at ±20 °C
5 (RTA)	Time	20 ± 3 s	98% at ±3 s; 90% at ±5 s
5 (RTA)	Atmosphere (O_2/H_2)	< 1 ppm	100% if < 1 ppm; 50% if 1–10 ppm; 10% if > 10 ppm
6 (Pillar etch)	CD	20 ± 2 nm	99% at ±2 nm; 85% at ±3 nm; 50% at ±4 nm
6 (Pillar etch)	Etch depth	14.82 ± 0.5 nm	99% at ±0.5 nm; 85% at ±1 nm
8 (CMP pillar)	Dishing	≤ 0.5 nm	99% at ≤0.5 nm; 85% at 1 nm; 60% at 2 nm
9 (Dry etch-back)	Anisotropy	≥ 3:1	99% at ≥3:1; 80% at 2:1; 0% (pillar collapse) at <2:1
11 (Cap CMP)	Al_2O_3 overlap	**≤ 1.0 nm**	100% at ≤1 nm; 20% at 1–2 nm; **0% at >2 nm** (hard cliff)
14 (Disk etch)	Sidewall roughness	< 1.5 nm RMS	99% at <1.5 nm; 85% at 2 nm; 60% at 3 nm
14 (Disk etch)	Conical angle	< 5°	99% at <5°; 80% at 10°; 40% at 20°

Step 11 is a hard cliff. The cap CMP is a hard yield cliff, not a soft window. If Al_2O_3 remains over the annular HZO field by more than 2 nm, the read signal is blocked and yield drops to zero. The 1.0 nm spec is the design target; devices between 1.0 and 2.0 nm overlap have degraded read margin but may pass on a per-wafer basis; devices above 2.0 nm are scrapped.

IV.5 Process Windows and Tolerances

Step	Parameter	Specification	Measurement
1	Buried oxide thickness	100 nm ± 5%	Ellipsometry
2	Gate length	28 nm ± 2 nm	CD-SEM
3	BE TiN thickness	50 nm ± 1 nm	Ellipsometry / XRF

Step	Parameter	Specification	Measurement
4	HZO thickness (as-deposited)	15.60 nm ± 1%	Ellipsometry (9 sites)
4	Hf:Zr ratio	50:50 ± 2%	XPS
5	RTA temperature	450 ± 10 °C	Thermocouple + pyrometer
5	RTA atmosphere	O_2, H_2 < 1 ppm	Inline mass spec
6	Pillar CD	20 nm ± 2 nm	CD-SEM
6	Pillar depth	14.82 ± 0.5 nm	OES endpoint + TEM on 1 wafer/ lot
7	TiN fill	Void-free (≥99% step coverage)	TEM on 10 cells
8	CMP dishing	≤ 0.5 nm	AFM
9	Etch-back depth	7 ± 0.5 nm	AFM
9	Anisotropy	≥ 3:1	TEM cross-section
10	Cap thickness	7 ± 1 nm	Ellipsometry
11	Cap CMP dishing	≤ 0.5 nm	AFM
11	Al_2O_3 overlap	≤ 1.0 nm	AFM (5 sites/wafer)
12	TE TiN thickness	50 nm ± 1 nm	Ellipsometry
13	TE CD	40 nm ± 2 nm	CD-SEM
14	Disk CD	40 nm ± 5%	CD-SEM
14	Sidewall roughness	< 1.5 nm RMS	AFM
14	Conical angle	< 5°	TEM
15	Post-etch anneal atmosphere	N_2 only	Inline mass spec (interlock)
17	Passivation thickness	100 ± 10 nm	Ellipsometry
18	Via CD	50 ± 5 nm	CD-SEM
19	Cu thickness	1000 nm ± 10%	XRF

IV.6 Yield Model

The projected yield is 77.7% on a 100 mm² die. The yield model uses a per-step yield mix reflecting the difficulty of each step, not a uniform yield assumption.

Yield per step	Number of steps	Cumulative
0.995 (near-perfect)	5	0.9752
0.99 (standard)	10	0.9044

Yield per step	Number of steps	Cumulative
0.985 (challenging)	5	0.9273
0.95 (critical — cap CMP)	1	0.95
Product (77.7%)	**21**	**0.7768**

Steps at each yield level. The 0.995 steps are: substrate prep, TE pattern, passivation, bond pad, Cu anneal. The 0.99 steps are: FinFET front-end, BE TiN, HZO ALD, TiN fill, post-etch anneal, post-TE anneal, via etch, Cu fill, final anneal, and one additional standard step. The 0.985 steps are: pillar hole etch, pillar CMP, disk etch, disk pattern (M7), via etch. The 0.95 step is: cap CMP (step 11, hard-cliff yield).

Wafer-level yield. 615 gross dies per 300 mm wafer × 77.7% = 478 good dies per wafer.

IV.7 Mask Set

Mask	Purpose	CD	Type
M2	Bottom electrode	160 nm	Dark
M3	Pillar hole pattern	20 nm	Dark
M4	Reserved (product-specific)	40 nm	Dark
M5	Top electrode	40 nm	Dark
M6	Reserved (product-specific)	80 nm	Dark
M7	Disk pattern	40 nm	Clear
M11	Extended-drain implant	200 nm	Dark

7 novel masks (M2, M3, M4, M5, M6, M7, M11). The pillar hole pattern uses 193 nm immersion lithography with LELE (Litho-Etch-Litho-Etch) to achieve the 20 nm CD.

IV.8 Packaging

Parameter	Value
Package type	BGA-324
Grid	18 × 18
Ball pitch	0.8 mm
Package size	19 mm × 19 mm
Package height	1.2 mm
Data rate	6.4 Gbps (single-ended DQS)
Die attach	Epoxy, 25 µm

Parameter	Value
Wire type	Gold, 25 µm
Bond pad pitch	100 µm
Package cost	$1.00 (volume)

IV.9 Thermal Analysis

State	Power	Thermal rise (θ_{JA} = 20 °C/W)	Junction at 85 °C ambient
Idle	109 mW	2.2 °C	87.2 °C
50% duty (read-sat)	1021 mW	20.4 °C	105.4 °C
100% duty (read-sat)	1934 mW	38.7 °C	**123.7 °C**
100% duty (write-sat)	~221 mW	4.4 °C	89.4 °C

Thermal rule. At 100% duty (read-saturated), the device requires either a thermal pad (θ_{JA} = 15 °C/W, giving junction 113.0 °C) or ambient derating to ≤80 °C. At ≤50% duty, the standard package is sufficient. At idle, thermal margins are comfortable.

IV.10 Power Delivery

Rail	Voltage	Purpose
VDD_CORE	1.0 V	Digital logic, comparator
VDD_PLL	1.0 V	Clock
VDDQ	1.8 V	Standard I/O (DQ, DQS, CA)
VDD_ANALOG	1.8 V	DAC, ADC, TIA, reference
VPP (inhibit)	2.5 V	Inhibit driver
VPROG (write)	3.5 V	Write driver, from on-die charge pump
V_WL (word-line boost)	1.5 V	SWL drive, from on-die charge pump

Power sequencing. Rails must come up in the order: VDD_CORE → VDD_PLL → VDDQ → VDD_ANALOG → VPP → VPROG → V_WL. Any order is acceptable within 1 ms except VPROG and V_WL, which must be last. All rails must reach 90% of nominal before reset is released. The charge pump start-up time is ~50 µs; VPROG is available for write commands after that delay.

IV.11 Equipment List

Equipment	Purpose

Equipment	Purpose
ALD (TiN, Al_2O_3, HfO_2, ZrO_2)	Bottom electrode, pillar fill, ferroelectric, cap, top electrode
CMP	Pillar planarization (step 8), cap planarization (step 11)
RTA furnace	Post-deposition anneal (step 5)
Immersion lithography (193 nm)	Pillar, top electrode, disk patterns
RIE etcher (Cl_2/Ar, Cl_2/BCl_3)	Pillar hole etch, disk etch, via etch
PECVD	Passivation deposition
PVD (Cu seed)	Cu seed layer
Electroplating	Cu interconnect fill
CMP (Cu)	Planarization after Cu fill
Optical emission spectroscopy	Pillar-hole endpoint (Ti 363.5 nm), disk endpoint (Si 251.6 nm)
In-situ interferometry (CMP)	Pillar and cap planarization endpoint
Dicing saw	Die separation
Wire bonder	Package assembly
XPS	Composition measurement (Hf:Zr ratio)
Spectroscopic ellipsometer	Film thickness measurement
GI-XRD	Phase fraction measurement
TEM (cross-section)	Pillar fill conformality, dry etch-back anisotropy
AFM	Cap CMP dishing and overlap measurement
Inline mass spec (RTA)	O_2/H_2 monitoring (hard-fail interlock)

IV.12 Foundry Requirements

The foundry must have:

- Standard 28 nm CMOS process with $CoSi_2$ source/drain (not NiSi).
- 10 V extended-drain option (standard in 28 nm CMOS).
- ALD capability for HfO_2/ZrO_2 (standard for high-k gate dielectric).
- ALD capability for Al_2O_3 with TMA + H_2O chemistry (standard).
- Post-CMOS ferroelectric anneal capability (RTA at 450 °C).
- CMP with in-situ optical endpoint (standard on advanced CMP tools).
- HZO ALD precursor handling ($HfCl_4$, $ZrCl_4$).
- HCl scrubber for the ALD exhaust.
- 193 nm immersion + LELE for the 20 nm pillar CD.

Foundry qualification. The $CoSi_2$ requirement and the HZO ferroelectric ALD capability narrow the foundry list. Qualified foundries are TSMC 28HPC+ and GlobalFoundries 22FDX.

Both are qualified for this product; if one becomes unavailable, the other can supply.

IV.13 Supply Chain

Material	Supplier category
$HfCl_4$ (99.99%)	ALD precursor supplier
$ZrCl_4$ (99.99%)	ALD precursor supplier
$TiCl_4$ (99.99%)	Standard semiconductor precursor
$Al(CH_3)_3$ (TMA)	Standard semiconductor precursor
NH_3 (99.999%)	Standard semiconductor gas
H_2O (18.2 MΩ·cm)	In-house ultrapure water
CMP slurry (silica)	Standard CMP supplier
300 mm Si (100) wafers	Standard wafer supplier
BGA-324 substrate	Package supplier
Gold bond wire (25 µm)	Standard
Die attach epoxy	Standard

IV.14 Production Volume and Lead Times

Parameter	Value
Good dies per wafer (615 gross)	478
Wafers per 100,000 units	209
Mask fabrication lead time	8–12 weeks
Wafer fabrication (28 nm CMOS + ferroelectric back-end + CMP)	12–14 weeks from mask release
Assembly and test	3–4 weeks
Total first-production lead time	~24 weeks
Steady-state lead time (masks available)	15–18 weeks

Part V — Circuit Design

V.1 Circuit Overview

The circuit architecture is built around the current-mode read path and the four-level write-verify ladder.

Key circuit subsystems:

- **Read path:** current-mode latch comparator with auto-zeroing, reference current generator, and read-voltage trim DAC.
- **Write path:** four-level H-bridge write driver, inhibit driver, and charge pump.
- **Hierarchical word-line architecture:** global row decoder, local row decoder, and sub-word-line (SWL) drivers.
- **Reference:** 60 kΩ resistor-based current reference, trim range ±30%.
- **ECC:** SECDED (72/64), internal, transparent to host.
- **Power management:** four power states (P0 active, P1 reduced clock, P2 clock-gated, P3 deep sleep).
- **Clock generation:** charge-pump PLL, 100 MHz to 1 GHz.
- **Temperature sensing:** 4 on-die sensors, max-of-4 + 3 °C margin reporting.

V.2 The Read Comparator

V.2.1 Circuit description

The comparator discriminates between the two resistance states by comparing the cell current to a reference current. It is a current-mode latch with four stages:

1. **Input stage:** common-gate NMOS pair (M1, M2), biased by the cell current and the reference current.
2. **Current mirror:** PMOS mirror (M3, M4) loads the input pair.
3. **Latch:** cross-coupled inverter pair (M5–M8) regenerates the output.
4. **Output buffer:** standard inverter (M9, M10).

The current-mode latch has no integration capacitor. The cell current and the reference current are compared directly at the input pair; the latch amplifies the difference. With the dielectric cap over the pillar, the cell current is the annular current only; the pillar-top shunt is blocked, so the comparator sees the full state-discriminating signal.

V.2.2 Transistor dimensions

Transistor	Type	W/L	Function
M1, M2	NMOS	100 µm / 28 nm	Input pair, offset 1.8 mV

Transistor	Type	W/L	Function
M3, M4	PMOS	5 µm / 28 nm	Current mirror, Z_{out} 100 kΩ
M5, M6	NMOS	2 µm / 28 nm	Latch pull-down, $g_m \approx 1$ mS
M7, M8	PMOS	2 µm / 28 nm	Latch pull-up, matched to M5/M6
M9, M10	CMOS	1 µm / 28 nm	Output buffer, 50 fF load

V.2.3 Comparator timing

Corner	Value
TT (nominal)	0.178 ns
SS (worst)	0.354 ns
Combined corner (SS + cold + low V + aged)	0.447 ns

V.2.4 Auto-zeroing

The comparator offset is canceled by auto-zeroing:

1. **Offset sampling:** the input pair is shorted (both inputs at the reference current). The offset is stored on a 100 fF capacitor at the latch input.
2. **Read:** the input pair is connected to the cell. The stored offset is subtracted from the signal.

Auto-zeroing is performed every 1,000 accesses. The offset drift over 1,000 accesses is negligible. The auto-zeroing is done in the background (no access stalls). The overhead is 5 ps every 1,000 accesses, about 0.0001% of the 4.71 ns serial read latency.

V.3 The Reference Current

V.3.1 Reference resistor

The reference current is set by a 60 kΩ resistor divider, trimmed during production test with a 4-bit programmable network (16 levels):

$$I_{ref} = V_{read} / R_{ref} = 9.88\ \text{mV} / 60\ \text{k}\Omega = 0.165\ \mu\text{A}$$

V.3.2 Resistor implementation

Recommended implementation: silicided poly (tempco ≤ −200 ppm/°C). This gives the largest corner margin. Unsilicided poly (tempco ∼−1500 ppm/°C) is permitted at reduced corner margin; the qualification report must document the corner SNR achieved with the chosen implementation.

V.3.3 Trim range and corner SNR

Parameter	Value
R_ref	60 kΩ
I_ref nominal	0.165 μA
Trim range	±30%, 0.115–0.215 μA
Trim resolution	4-bit (16 levels)
Corner SNR (silicided poly)	3,733×
Corner SNR (unsilicided poly)	~1,500×

V.3.4 Reference trip point

The reference current (0.165 μA) is set at approximately 15% of the on-state cell current (1.127 μA). The off-state leakage floor is far below the trip point (~10 fA vs 0.165 μA), so the off-state decision is trivially safe. The large positive differential (0.962 μA) gives fast on-state comparator decisions (0.354 ns at SS corner).

V.4 Read Voltage Trim DAC

Parameter	Value
Resolution	4-bit (16 levels)
Range	8.5–14.0 mV
Step size	~0.37 mV (5.5 mV / 15 intervals)
Power	16.6 nW
Area	0.001 mm²
Number	1 (global, per die)

The trim value is set at production test based on the measured coercive field of the die. It is stored in a 4-bit OTP register with 4× redundancy and loaded at boot. A separate temperature-tracking lookup table (76 entries, 1 °C resolution) adjusts V_read for temperature; the trim DAC and the LUT together cover the full operating temperature range.

Max trim limited to 14.0 mV. At 16.0 mV (the previous maximum), the read-disturb margin at 85 °C drops to 12.4× — above the 10× requirement but only by 24%. At 14.0 mV, the margin is 14.2× (42% above the requirement). The maximum trim is 14.0 mV for this design.

V.5 Hierarchical Row Decoder and Sub-Word-Line Drivers

V.5.1 Architecture

Each 1,024-cell main word line is divided into 2 sub-word-lines (SWLs) of 512 cells each, center-fed. The main word line enable is driven from a global row decoder that serves a group of 8 sub-arrays, then amplified by a local row decoder at each sub-array.

V.5.2 Word-line IR drop

Parameter	Value
Cells per active SWL segment	512
Sub-word-line current per segment	289 µA
Segment resistance	1.088 kΩ
IR drop per segment	315 mV
V_WL boost rail	1.5 V

The V_WL boost rail is provided by a small on-die charge pump. The 512-cell SWL is center-fed, so the current splits left and right and the per-half IR drop is one-quarter of the single-end lumped value.

V.5.3 Sub-word-line drivers

Parameter	Value
Type	Source follower (NMOS)
Transistor size	10 µm / 28 nm
Quiescent current	10 µA per active SWL driver
Power per active driver	10 µW
Active drivers at any time	16 SWL segments
Active power per bank	160 µW
Number of active banks	Up to 4 simultaneously

V.6 The Write Driver

V.6.1 Circuit

Parameter	Value
Topology	H-bridge
Voltage range (production)	−2.890 to +2.890 V
Rise time	300 ps
Current capability	10 µA per cell

Parameter	Value
Transistor type	5 V I/O (28 nm CMOS)
Ladder levels	2.223 / 2.445 / 2.668 / 2.890 V

The maximum write voltage is 2.890 V, within the 5 V I/O limit. No cascode is needed for write. The H-bridge reverses polarity for the erase operation; no separate negative charge pump is required.

V.6.2 Write-verify control

Each write operation consists of up to four pulse-verify cycles. After each pulse, the word is read back and verified. If all cells have switched, the write completes. If not, the next ladder level is applied. The write-verify controller is a small state machine inside the write driver column logic.

Parameter	Value
Pulse width	10 ns
Verify time	2 ns
Setup time	1.5 ns
Max attempts	4
Average attempts (per word)	3.64
Write latency (avg)	45.2 ns
Write latency (max)	49.5 ns

V.6.3 Cap field during write

At the highest write voltage (2.890 V), the field across the 7 nm Al_2O_3 cap is 4.13 MV/cm — below the Al_2O_3 dielectric breakdown strength (5–10 MV/cm) with 1.2–2.4× margin.

V.7 The Inhibit Driver

Parameter	Value
Topology	Source follower
Transistor type	2.5 V I/O NMOS
Voltage	0.95 V ± 5%
Rail	VPP = 2.5 V (resistive divider)
Power	Active only during programming

Inhibit voltage 0.95 V. The inhibit voltage is set at 0.95 V, giving an inhibit field of 0.641 MV/

cm across the 14.82 nm ferroelectric film. This is 47.9% of the coercive field at 85 °C — below the standard 50% threshold for reliable no-disturb. Unselected cells on the same word line as a written cell see 47.9% of E_c and do not switch.

V.8 Charge Pumps

V.8.1 Write pump

Parameter	Value
Topology	Dickson, 7-stage positive
Sustained current	80 mA (4-bank parallel)
Peak current per word	520 mA (bitline-limited, absorbed by cell capacitance)
Output	+3.5 V positive (used for both write and erase via H-bridge polarity reversal)
Efficiency	~75%
Area	~0.4 mm²
Decoupling	500 pF per bitline group
Ripple	≤ 170 mV (absorbed by local decoupling)

No negative charge pump is required. The H-bridge reverses polarity between write (top electrode positive) and erase (bottom electrode positive). The pump generates only the positive +3.5 V output.

Sustained vs peak current. The sustained current (80 mA) is what the pump must deliver over a burst of word writes. The peak per-word current (520 mA) is the local recharge current for 64 bitlines × 1.836 pF each, from 0 to V_write in ~1 ns. This peak is absorbed by the bitline capacitance and does not stress the pump. Local decoupling at the bitline drivers (500 pF per group) rides through the 1 ns edge. The pump recharges the decoupling between bursts at the sustained rate.

V.8.2 V_WL boost pump

Parameter	Value
Output	1.5 V
Power	0.2 mW
Area	0.02 mm²

V.9 Level Shifter and Voltage Reference

V.9.1 Level shifter

Parameter	Value
Input	1.0 V logic
Output	2.5 V gate drive
Delay	100 ps
Area per shifter	50 µm²

V.9.2 Voltage reference

Parameter	Value
Type	Bandgap reference
Accuracy	±1%
Temperature coefficient	< 50 ppm/°C
Power	100 µW

V.10 Bank Architecture

V.10.1 Top-level hierarchy

Level	Value
Banks per die	16
Sub-arrays per bank	930
Rows per sub-array	1,024
Cols per sub-array	1,024
SWL length	512 cells (center-fed)
SWL segments per MWL	2
Cells per die (raw)	15.6 Gb
Active banks (max)	4 simultaneously

V.10.2 Address map

Field	Bits
Column (within 64-bit word)	6 bits
Column word (within row)	4 bits (16 words)
Row	10 bits
Sub-array (within bank)	10 bits

Field	Bits
Bank	4 bits
Total address bits	**34**

V.10.3 Decoder power

Decoder	Specification	Power
Global row decoder	14-bit input, drives row-select to 8 sub-arrays	5 mW
Local row decoder	10-bit input, drives main word line	8 mW
Sub-word-line decoder	4-bit input, drives SWL segment pass gates	2 mW
Column decoder	10-bit column address, drives sense-amp select	5 mW
Total decoder power	—	**20 mW**

V.11 ECC

Parameter	Value
Code type	SECDED (single-error-correct, double-error-detect)
Data word	64 bits
Check bits	8 bits (64 → 72 encoding)
Code rate	0.875
Overhead	12.5%
Correctable errors	1 bit per 64-bit word
Detectable errors	2 bits per 64-bit word
Power	11.3 mW

SECDED is required to handle the random bit-failure rate from cycling. The 8 check bits are internal to the die and transparent to the host; the external data bus is 64-bit.

V.12 Bad-Block Management

1. At production test, identify bad blocks (blocks with uncorrectable errors after the wake-up cycle).
2. Store the bad-block list in a reserved area (block 0, 2% of the total block count).
3. During operation, skip bad blocks.
4. Periodically scan for new bad blocks (background, low priority).

V.13 Power Management States

State	Description	Power	Wake time
P0	Full operation	~109 mW	—
P1	Reduced clock (50%)	~70 mW	10 ns
P2	Idle (clock gated)	~30 mW	10 ns
P3	Deep sleep (VPROG off, data non-volatile)	1–5 mW	50 µs (VPROG restart)

The data is non-volatile, so power-off is instant and lossless. Power-on restores the device with no data loss. The P3 to P0 wake time is 50 µs because the charge pump must restart.

V.14 Clock Generation

Parameter	Value
Reference	100 MHz
PLL type	Charge-pump PLL
Output frequency	100 MHz to 1 GHz
Jitter	< 1 ps RMS
Lock time	< 10 µs
Power	10 mW

V.15 Temperature Sensor

Parameter	Value
Type	Bandgap-based
Accuracy	±1 °C per sensor
Resolution	0.25 °C
Power	1 µW
Area	0.01 mm²
Conversion time	10 µs
Number of sensors	4 per die
Placement	Row decoder, I/O drivers, sense amplifier array, reference array
Reported temperature	Max of 4 sensors + 3 °C safety margin

Max-of-4 with 3 °C margin. The four sensors are placed at the row decoder (hot spot), I/O drivers (hot spot), sense amplifier array (warm), and reference array (cool). The reported

junction temperature is the maximum of the four, plus 3 °C for sensor placement error and self-heating not captured by the sensor.

V.16 Design-for-Test

V.16.1 JTAG Test Access Port

Signal	Function
TCK	Test clock
TMS	Test mode select
TDI	Test data input
TDO	Test data output
TRST (optional)	Test reset (on-die pull-up)

V.16.2 Built-in self-test

- Pattern generator: March C- pattern for burn-in; March 17N for final test.
- Comparator: compares read data to expected data.
- BIST result: pass/fail flag.
- BIST time: full array test at the two-mode test protocol (Part VII).

V.16.3 Redundancy and repair

Each sub-array includes 2 spare rows and 2 spare columns (0.39% spare fraction), programmed by laser fuse or e-fuse. The repair is applied at production test based on the BIST result.

V.17 VPROG ESD Protection

Parameter	Value
Type	Dedicated clamp (diode + resistor)
Rail	VPROG (3.5 V)
Rating	500 V CDM
Trigger	Local clamp, activates on transient overshoot

The VPROG rail includes a dedicated ESD clamp rated to 500 V CDM. The clamp is a standard cell (diode + resistor), local to the VPROG pin.

V.18 Complete Circuit Power Summary

Component	Power
Comparators (sense amps)	1.16 mW
Word-line drivers + decoders	10 mW
I/O on-chip driver (idle)	50 mW
Clock (PLL + distribution)	10 mW
ECC (SECDED)	11.3 mW
Charge pumps (write + V_WL)	0.5 mW
Leakage at 87 °C	6 mW
Control logic	20 mW
Write BL settling (avg)	0.02 mW
Total internal (idle)	**109 mW**
I/O dynamic at 50% duty	+912 mW
I/O dynamic at 100% duty	+1825 mW

Per-bank read decoupling: 10 nF. Per-bitline-group write decoupling: 500 pF. All values at Tj = 87 °C (steady state, 85 °C ambient + thermal rise).

Part VI — Operating Principles

VI.1 Overview

The Vortex FeRAM supports three fundamental operations: write, read, and erase. Write and erase use the four-level write-verify ladder with opposite polarities. Read uses the current-mode latch comparator with the reference current at 0.165 μA. All operations use the hierarchical word-line architecture (512-cell sub-word-lines, center-fed).

VI.2 Write Operation

VI.2.1 Write procedure

For each 64-bit data word:

1. Decode bank address (1.0 ns).
2. Decode sub-array, row, and word address (0.5 ns).
3. Activate main word line and target sub-word-line driver (0.1 ns).
4. Apply write voltage attempt 1 (10 ns pulse).
5. Verify (read back, 2 ns).
6. If any cell in the word failed, apply attempt 2 (10 ns).
7. Verify (2 ns).
8. If any cell failed, apply attempt 3 (10 ns).
9. Verify (2 ns).
10. If any cell failed, apply attempt 4 (10 ns).
11. Verify (2 ns).
12. Cells that fail all four attempts are marked and repaired by ECC.

Average write time: 45.2 ns per 64-bit word. Maximum (all four attempts): 49.5 ns.

VI.2.2 Write voltage levels

Attempt	Field	Voltage	Cumulative switch fraction
1	1.50 MV/cm	2.223 V	0.500
2	1.65 MV/cm	2.445 V	0.875
3	1.80 MV/cm	2.668 V	0.984
4	1.95 MV/cm	2.890 V	1.000

All write voltages are within the 5 V I/O rail. No cascode is required for write.

VI.2.3 Per-word write distribution

The write-verify operates on a full 64-bit word at a time. A word requires all 64 of its cells to switch; the number of attempts needed is the maximum over the 64 cells.

Attempts needed	Probability
1	$< 10^{-19}$
2	~0.02%
3	~35.5%
4	~64.4%

Average: 3.64 attempts per word. This is why the average write latency (45.2 ns) is much higher than the per-cell average would suggest.

VI.2.4 Write energy

Parameter	Value
Write energy per cell (avg, 25 °C)	0.88 fJ
Write energy per cell (cold corner)	0.94 fJ
System write energy (including I/O)	~6.5 pJ per bit

The cell-level write energy is 0.88 fJ. System-level write energy including I/O is dominated by the I/O charge on the external bus (88 active pins × 4 pF × 1.8 V² × 0.5 activity).

VI.2.5 Cold-corner write behavior

At 10 °C and −5% VDD, the coercive field rises to 1.531 MV/cm, and level 1 of the ladder (2.112 V at −5% VDD) gives only 1.425 MV/cm — below $E_c(10\ °C)$. Cells do not switch at level 1 at the cold corner. They need level 2 or higher.

At the cold corner, every word needs all four attempts. Cold-corner write latency is **49.5 ns** (all four attempts). Cold-corner write energy is **0.94 fJ per cell**.

Writes below 0 °C are outside the validated operating range. The controller must warm the device to ≥10 °C before accepting writes. Reads are functional at all storage temperatures (−40 to 125 °C).

VI.3 Read Operation

VI.3.1 Read procedure

For each 64-bit data word:

1. Decode bank address (1.0 ns).
2. Decode sub-array, row, and word address (0.5 ns).
3. Activate main word line and target sub-word-line driver (0.1 ns).

4. Apply V_read to the top electrode (9.88 mV nominal).
5. Cell settling (0.18 ps, negligible).
6. Comparator decision (0.178 ns TT / 0.354 ns SS).
7. Data output through the output register and I/O driver (2.0 ns).

Total read latency (TT, serial first-access): 4.71 ns on-chip. Steady-state (pipelined, row already open): 2.63 ns on-chip. SS corner: 4.88 ns serial, 2.80 ns pipelined. System (add 3–5 ns controller): ~7.7–9.7 ns serial, ~5.6–7.6 ns pipelined.

VI.3.2 Read-disturb margin

Condition	Margin
Nominal V_read, 25 °C	22.4×
Nominal V_read, 85 °C	20.1×
Max trim (14.0 mV), 25 °C	15.8×
Max trim (14.0 mV), 85 °C	14.2×

All conditions above the 10× design requirement. The read field is far below the coercive field; the polarization state does not switch during a read.

VI.3.3 Read operation over 10^{12} reads

The read-disturb margin is a design constant over device lifetime. At 10^{12} reads, the accumulated read field exposure does not cause switching. For extended read life, the design supports 10^{12} reads at 25 °C and 10^{11} reads at 85 °C without degradation.

A secondary mechanism — imprint from repeated reads at one polarity — is under investigation. The device supports an optional SWAP command (Part IX) that inverts the polarity encoding for blocks that see asymmetric read/write patterns.

VI.4 Erase Operation

VI.4.1 Erase procedure

The erase operation writes the logical "0" state (winding number n = 0) to the cell. It uses the same write-verify ladder but with **opposite polarity**:

- Write operation: top electrode driven positive relative to bottom electrode, at the four ladder voltages.
- Erase operation: bottom electrode driven positive relative to top electrode, at the four ladder voltages.

The H-bridge reverses the polarity. **No separate negative charge pump is required**; the on-die positive pump output is used, and the H-bridge determines which terminal is positive.

VI.4.2 Erase latency and energy

Parameter	Value
Erase latency	Same as write: 45.2 ns avg / 49.5 ns max
Erase energy	±30% of write energy (asymmetric P-E loop)
Erase energy per cell	0.6–1.1 fJ (nominal 0.88 fJ ± 30%)

The asymmetric P-E loop of HZO means the erase energy is typically 10–30% lower or higher than the write energy, depending on the film's specific loop shape. The design uses 0.88 fJ as the nominal for both directions; the customer should expect ±30% variation.

VI.5 Temperature Compensation

VI.5.1 Temperature dependence of the coercive field

The coercive field decreases with temperature. The design uses the following interpolation for the write-voltage lookup table:

Temperature	E_c	Ratio to 25 °C
10 °C	1.531 MV/cm	1.026
25 °C	1.4923 MV/cm	1.000
55 °C	1.4141 MV/cm	0.948
85 °C	1.3377 MV/cm	0.896

The write-voltage lookup table is populated at production test from measured P-E loops at these four temperatures. The device works regardless of the specific temperature coefficient; the LUT is loaded from characterization, not computed from a formula.

VI.5.2 The write-voltage LUT

Parameter	Value
Temperature range	10–85 °C
Resolution	1 °C
Table size	76 entries × 4 ladder levels = 304 entries
Storage	On-die ROM (608 bytes)

The table is loaded at factory test and read at boot. It is not user-programmable.

VI.6 Periodic Recalibration

VI.6.1 Reference cell refresh

The memory uses dedicated reference cells (1,024 per bank in a separate column) programmed to known states. Two refresh mechanisms are implemented:

1. **Timer-based:** refresh every 10^5 writes per bank. For the Vortex FeRAM (1.5×10^8 writes per bank over lifetime), this triggers 1,500 times.
2. **Error-rate-triggered:** when the single-bit ECC error rate per bank rises above 2× baseline, issue a refresh. This catches workload imbalance where some banks see more writes than others.

The refresh procedure re-writes the reference cells to n=1, restoring the trip-point current. Refresh cost is 1% of write bandwidth, scheduled in the background.

VI.6.2 DQS calibration

The read path uses DQS (data strobe) for timing. DQS calibration is performed at boot and whenever the temperature changes by more than 10 °C. The calibration sweeps the DQS delay over ±500 ps in 5 ps steps (100 points), finding the center of the read data eye.

Calibration step	Duration
Write leveling	~200 µs
Read gate training	~300 µs
Read data eye training	~1 ms
Write data eye training	~1 ms
Vref training	~300 µs
Total per calibration	**~2.8 ms**

VI.7 Power-On and Power-Down Sequences

VI.7.1 Power-on sequence

1. Apply VDD_CORE (1.0 V) first.
2. Apply VDD_PLL (1.0 V) within 100 µs.
3. Apply VDDQ (1.8 V), VDD_ANALOG (1.8 V), and VPP (2.5 V) within 1 ms.
4. Apply VPROG (3.5 V) after VDD and VDDQ have stabilized. Wait for the charge pump soft-start and the output to reach its target (~50 µs).
5. Apply V_WL (1.5 V) last.
6. Wait for power-on reset (1 µs).
7. Load calibration data from on-die ROM.
8. Release reset.
9. Begin normal operation.

Total boot time: ~55 µs (VPROG charge pump is the dominant delay). First write command is

available after the boot delay.

VI.7.2 Power-down sequence

1. Complete pending operations.
2. Disable V_WL.
3. Disable the charge pump.
4. Remove VPROG.
5. Remove VPP.
6. Remove VDD_ANALOG, VDDQ.
7. Remove VDD_PLL, VDD_CORE.

The data is non-volatile, so power-off is lossless. No data is lost on power-down.

VI.7.3 Hot-plug behavior

If VPROG is present and VDDQ is off or below 90% of nominal, standard I/O pins could see VPROG through the write driver and be damaged. The device must not be hot-plugged with VPROG live and VDDQ off. The power-on reset monitors all rails and disables VPROG output until VDD and VDDQ are both up. A hot-plug detection circuit pulls VPROG to 0 V during any rail transition where VDDQ is not within 90% of nominal.

VI.8 Error Handling

VI.8.1 Correctable errors (single-bit)

1. ECC decodes the 72-bit word and corrects the single-bit error.
2. The error event is logged.
3. Operation continues without host intervention.

VI.8.2 Detectable errors (double-bit)

1. ECC detects the double-bit error (SECDED can detect but not correct).
2. The operation is retried from the host.
3. If the retry succeeds, the error is logged and operation continues.
4. If the retry fails, the block is marked as bad and the data is recovered from a redundant block.

VI.8.3 Uncorrectable errors (triple-bit or more)

1. ECC cannot correct or reliably detect.
2. The block is marked as bad.
3. The host is notified (ALERT_N).
4. Data recovery from redundant blocks or from host-side redundancy.

VI.9 The Wake-Up Procedure

All HZO films require a wake-up cycle before reaching full remanent polarization. The wake-up is performed once at production test:

1. Apply 10^4 write-verify cycles to all cells.
2. Measure the post-wake-up cell resistance and record the value in on-die OTP.
3. Update the calibration values used by the read and write paths.

Wake-up time per cell: $10^4 \times 21.2$ ns = 212 µs. After wake-up, the remanent polarization increases by 1.5–2×. The post-wake-up P_s is the value stated in this document (13.07 µC/cm²).

Production test cost. The wake-up cycle adds ~30 seconds to the per-die production test time (parallel wake-up across all cells). This is included in the packaged test cost.

VI.10 Read-Disturb and Write-Disturb

VI.10.1 Read-disturb

The read field is sub-threshold for HZO. The polarization state does not switch during a read. The read-disturb margin against the vortex destroy field is 20.1× at 85 °C (worst case). The read field is well below the coercive field; no cumulative switching occurs.

VI.10.2 Write-disturb

The unselected cells on the same word line receive an inhibit voltage of 0.95 V. The net field across the unselected cell is 47.9% of E_c at 85 °C — below the 50% threshold for reliable no-disturb. The unselected cells are not disturbed during a write to an adjacent cell.

VI.11 Temperature Range and Cold-Start

Condition	Behavior
10–85 °C (validated)	Full operation (read and write)
0–10 °C (marginal)	Writes use extended ladder (levels 2–4). Cold-corner write latency 49.5 ns.
−40 to 0 °C (storage only)	Reads functional. Writes require warm-up to ≥10 °C.
−40 to 125 °C (storage range)	Data retained. No degradation during storage.

VI.12 Refresh Policy

The memory is non-volatile. No refresh is required during normal operation. The retention is > 10 years at 85 °C. For applications with extreme retention requirements, the customer can schedule a periodic refresh (re-write of stored data) at any interval shorter than the retention

specification.

VI.13 The Complete Operating State Machine

State	Entry	Exit	Power
Power-Off	VDD removed	Power-on sequence	0
Power-On Reset	All rails up	Reset released + calibration loaded	~50 mW (during boot)
P0 (Active)	Reset released	Idle timeout or software P2/P3 command	109 mW (idle)
P1 (Reduced Clock)	Software command	Access or software P0 command	~70 mW
P2 (Clock Gated)	Idle timeout or software command	Access or software P0/P3 command	~30 mW
P3 (Deep Sleep)	Software command	Access (50 µs wake) or software P0 command	1–5 mW

VI.14 The SWAP Command (Polarity Swap)

VI.14.1 What SWAP does

The memory cell encoding is: n=1 for "written one state," n=0 for "written zero state." If a block sees asymmetric cycling (e.g., 80% of writes are to n=1, 20% to n=0), the n=1 cells fatigue faster than the n=0 cells. Over time, this creates imprint: the n=1 cells have lower remanent polarization than the n=0 cells, degrading the read margin for the n=1 state.

The SWAP command inverts the encoding for a block: n=1 becomes "written zero state," n=0 becomes "written one state." After SWAP, the previously over-cycled n=1 cells are now written as n=0 and vice versa. This equalizes the cycling across the two polarization states.

VI.14.2 SWAP protocol

1. Each 4 KB block has a 32-bit write counter.
2. When the counter reaches 10^5, the controller may issue a SWAP command to the block.
3. The memory reads the block, inverts the polarity encoding, writes back, and resets the block's 1-bit polarity flag.
4. The counter resets to zero.

SWAP is **optional**. It is recommended for workloads with asymmetric read/write patterns (e.g., write-once, read-many applications). For workloads with symmetric cycling, SWAP is not required.

SWAP storage cost (when enabled): 1 bit per 64-bit word = 1.56% storage overhead.

VI.15 The Complete Operating Parameters

Parameter	Value
Read latency (system, serial)	~7.7–9.7 ns
Read latency (system, pipelined)	~5.6–7.6 ns
Write latency (system)	~48–50 ns (nominal); ~52 ns (cold corner)
Erase latency	Same as write
Read bandwidth	51.2 GB/s
Write bandwidth (pure-write)	0.7 GB/s
Read energy	42 aJ per bit
Write energy per cell	0.88 fJ (nominal); 0.94 fJ (cold corner)
Idle power (Tj = 87 °C)	109 mW
50% duty power (read-sat)	1021 mW
100% duty power (read-sat)	1934 mW
Thermal (100% duty, standard package)	Requires thermal pad (θ_{JA} = 15 °C/W) or ambient ≤80 °C
Endurance (1% fail, Tj = 87.2 °C)	2.45×10^9 cycles
Retention at 85 °C	> 10 years
Wake-up cycles	10^4
Operating temperature (full write)	10–85 °C
Operating temperature (read only)	−40 to 125 °C

Part VII — Verification and Reliability

VII.1 Overview

Verification is organized into three sequential gates plus full-chip characterization. Each gate catches a specific class of problems at the earliest possible stage. The gates are cumulative: if Gate 1 fails, Gate 2 is not attempted.

Gate	Purpose	When
Gate 1	Ferroelectric phase confirmation	After RTA anneal, before further patterning
Gate 2	Single-cell electrical characterization	After disk etch and post-etch anneal
Gate 3	Small-array verification	After full fabrication, before dicing
Full chip	Functional and performance verification	After dicing and packaging

Ship gates. The Vortex FeRAM ships contingent on four specific measurements. If any of the four fails outside its acceptance range, production is held until resolution. The four ship gates are:

1. **Core DC conductivity** (Gate 2, M1). The read mechanism requires the ferroelectric core to be conductive. If measured cell current is below 0.1× the design value, the read path is broken.
2. **Core radius** (Gate 2, M1). The design value is 2.317 nm. If measured r_c exceeds 3.01 nm (30% above the design value), the annular/r_c ratio drops from 4.32 to 3.32 — still comfortable, but the design window narrows. A hard r_c ship gate of 5.0 nm (annular/r_c = 2.0) applies; beyond that the design margin is not sufficient.
3. **Dry etch-back anisotropy** (process qualification). The pillar etch-back must achieve ≥ 3:1 anisotropy. Below 3:1, the pillar begins to shrink and the cell resistance rises out of spec.
4. **CMP endpoint** (process qualification). The cap CMP must stop at the HZO field plane with ≤ 1.0 nm Al_2O_3 overlap over the annular region. Above 1.0 nm, the read path is partially blocked.

VII.2 Gate 1 — Ferroelectric Phase Confirmation

VII.2.1 Test structure and procedure

Test structure: A 50 μm × 50 μm capacitor in the scribe line or a dedicated test area, with the TiN/HZO/TiN stack.

Procedure:

1. Contact the top and bottom electrodes with probe tips.
2. Apply a voltage sweep from −5 V to +5 V and back.

3. Measure the polarization as a function of voltage (P–V loop).

VII.2.2 Pass criteria

Measurement	Target	Pass criterion
Hysteresis loop	Open loop	Clearly visible
Remanent polarization P_s	> 10 μC/cm²	Above threshold (design 13.07 μC/cm²)
Coercive field E_c	1.0–2.0 MV/cm	Within range (design 1.4923 MV/cm)
Film thickness	14.82 nm ± 1%	Ellipsometry (9 sites)
Orthorhombic phase fraction	50–70%	GI-XRD

VII.2.3 If Gate 1 fails

Symptom	Likely cause	Fix
No hysteresis	Film not crystallized	Increase RTA temperature to 500 °C
Low P_s	Low orthorhombic fraction; pre-wake-up	Increase RTA time; run wake-up cycle
Wrong coercive field	Wrong film thickness or Hf:Zr composition	Check ALD cycle count and XPS ratio
Leaky capacitor	Pinholes	Check contamination; verify ALD chamber cleanliness
RTA atmosphere excursion	O_2 or H_2 > 1 ppm during anneal	Gas-line interlock should have caught this; investigate interlock failure

VII.3 Gate 2 — Single-Cell Electrical Characterization

VII.3.1 Test structure and procedure

Gate 2 uses individual memory cells fabricated on the same wafer as the production die, or on a dedicated test die with the same process flow. Seven measurements (M1–M7) are performed on 12 wafers (4 per measurement for redundancy). All seven measurements must pass before proceeding to Gate 3.

VII.3.2 M1 — Cell I-V vs disk diameter

Setup: Four disk diameters (30, 40, 50, 60 nm). Each disk measured at V_{read} = 9.88 mV. Pillar diameter and film thickness fixed.

Measurement: I_{read} at n=1 for each diameter. Plot I_{read} vs disk area.

Pass criterion: The measured I_{read} at the design disk diameter (40 nm) is within a factor of 3 of the design value (1.127 μA). The measured cell current at the four diameters is consistent

with one of the standard conduction models (annular-only, full-disk, pillar-parallel, core-only) within 10% across all four diameters.

Ship gate: If I_read at the design disk diameter is below 0.1× the design value (< 0.113 μA), the ferroelectric core is not conductive and the read path is broken. Redesign required (6–12 months).

VII.3.3 M2 — P–E loop and loop area

Setup: 50 μm × 50 μm HZO capacitor (same as Gate 1).

Measurement: Full P–E loop at 25 °C. Extract P_s, E_c, and loop area (the closed-loop area in J/m³).

Pass criterion: P_s within 10% of 13.07 μC/cm²; E_c within 5% of 1.4923 MV/cm; loop area in 2.0–3.5 × (P_s · E_c).

Use: The loop area sets the per-cell switching energy. The design value of 0.88 fJ per cell assumes a loop area of 2.49 × (P_s · E_c). If the measured loop area is 3.5 × (P_s · E_c), the per-cell write energy rises to 1.24 fJ, and the write pump current spec (80 mA sustained) is re-qualified.

VII.3.4 M3 — Above-pillar leakage

Setup: Test structure with the annular path blocked (top-electrode geometry exposes only the pillar top).

Measurement: I-V at V_read = 9.88 mV.

Pass criterion: Measured current < 1 pA. The 7 nm Al_2O_3 cap blocks DC current through the pillar-top region.

If fails: Cap deposition or CMP issue. The Al_2O_3 cap is not hermetic, or the CMP under-polished leaving a thinner than 6.5 nm effective cap. Requires deposition recipe or CMP endpoint investigation.

VII.3.5 M4 — E_c vs temperature

Setup: 50 μm × 50 μm capacitors at 10, 25, 55, 85 °C.

Measurement: P–E loop at each temperature. Extract E_c(T).

Pass criterion: E_c(T) matches the linear interpolation E_c(25 °C) × [1 − (T − 25)/579.65] within 5%. The measured temperature coefficient is used to populate the write-voltage lookup table.

VII.3.6 M5 — Switching distribution

Setup: 1,000 individual cells. Pulse amplitude sweep at fixed 10 ns pulse width.

Measurement: Fraction of cells switched vs pulse amplitude. Extract the distribution width σ.

Pass criterion: Distribution width $\sigma \leq 0.20$ (consistent with the four-level ladder in Part VI). If σ is between 0.20 and 0.25, the write path uses a wider ladder or an extra verification step. If $\sigma > 0.25$, the write path is unreliable (grain structure investigation required).

VII.3.7 M6 — Retention projection

Setup: 256 individual cells. Programmed to n=1, post-wake-up.

Measurement: Bake at 125, 150, 175, 200 °C for 24, 100, 500, 1000 hr. Measure cell resistance after each bake.

Pass criterion: Extrapolated 10-year retention at 85 °C with 2× margin (i.e., measured retention ≥ 20 years equivalent at 85 °C).

VII.3.8 M7 — Extended endurance

Setup: 64 cells per material, pre-screened to require 4 attempts (i.e., cells that fail to switch at levels 1, 2, 3).

Measurement: Apply 4-attempt cycles continuously. Measure P_s at 10^4, 10^5, 10^6, 10^7 cycles.

Pass criterion: Extrapolated endurance (1% fail) is consistent with the design spec of 2.45×10^9 cycles. If endurance is measured lower than 1×10^7 cycles, the endurance spec drops to the measured value (still 100× NAND SLC, still endurance-class), and the tier positioning is updated.

VII.3.9 Gate 2 decision tree

M1 (I-V)	M3 (cap)	M5 (σ)	M6 (retention)	M7 (endurance)	Outcome
Pass	Pass	Pass	Pass	Pass	**Gate 2 PASS** — proceed to Gate 3
Pass	Pass	Pass	Pass	Below 1e7	Endurance spec drops to measured; ship with updated spec
Pass	Fail	Any	Any	Any	Cap deposition issue; re-qualify Al_2O_3 ALD
Pass	Pass	$\sigma > 0.25$	Any	Any	Grain structure investigation; ALD recipe change
Pass	Pass	Pass	Fail	Any	Retention not adequate; interface / pinning investigation
Fail	Any	Any	Any	Any	**Gate 2 FAIL** — 6–12 month redesign

VII.4 Gate 3 — Small-Array Verification

VII.4.1 Test structure and procedure

Gate 3 uses 1M-cell arrays fabricated on 10 wafers per lot. Two test modes:

- **Production mode:** 10,000 cells sampled per die, run at final packaged test on every die. Tests basic functionality and yield.
- **Qualification mode:** 1,000,000 cells per die, run on 10 wafers per lot. Tests defect density, cell-to-cell variation, and full-array functionality.

VII.4.2 Pass criteria

Measurement	Target
Array yield (production mode)	> 80%
Array yield (qualification mode)	> 80%
Cell-to-cell variation (1σ)	< 20%
BIST March 17N	Pass

VII.5 Intermediate Tests

Step	Measurement	Target	Method
After HZO deposition	Film thickness	±1%	Ellipsometry
After HZO deposition	Hf:Zr composition	50:50 ± 2%	XPS
After CMP (step 8)	Pillar-top exposure, planarity	Full exposure, < 0.5 nm	AFM + SEM
After cap CMP (step 11)	Al_2O_3 overlap over HZO field	≤ 1.0 nm	AFM (5 sites)
After anneal	Phase fraction	50–70% orthorhombic	GI-XRD
After pillar etch (step 6)	Pillar CD	20 nm ± 2 nm	CD-SEM
After pillar dry etch-back (step 9)	Anisotropy	≥ 3:1	TEM cross-section
After disk etch (step 14)	Disk CD	40 nm ± 5%	CD-SEM
After disk etch	Edge roughness	< 1.5 nm RMS	AFM
After passivation	Film stress	< 200 MPa	Wafer bow

VII.6 Full-Chip Characterization

After packaging, every die undergoes full-chip characterization:

Test	Pass criterion

Test	Pass criterion
Full array functional test (March 17N)	All cells read and write correctly; ECC single-bit errors recoverable; no uncorrectable errors
Read latency (on-chip, TT, serial)	4.71 ns ± 10%
Read latency (on-chip, TT, pipelined)	2.63 ns ± 10%
Read latency (on-chip, SS, serial)	4.88 ns ± 10%
Write latency (per word, all corners)	45.2 ns avg ± 5%; 49.5 ns max ± 2%
Read-disturb margin	≥ 15× at 85 °C, max trim
Idle power (Tj = 87 °C)	109 mW ± 10%
50% duty power (read-saturated)	1021 mW ± 15%
100% duty power (read-saturated)	1934 mW ± 15%
Thermal rise (100% duty, standard package)	≤ 40 °C at 20 °C/W
DQS training (all lanes)	Pass
JTAG IDCODE	Correct
OTP calibration load	Pass

VII.7 Reliability Test Matrix

Test	Condition	Duration	Pass criterion
HTOL (endurance)	85 °C, 1.1× VDD, 1 Hz cycling	1000 h (3.6×10^6 cycles)	> 1×10^9 cycles equivalent (endurance spec)
HTOL (retention)	125 °C, 1.1× VDD, no cycling	1000 h	> 10-year equivalent at 85 °C
HTSL	125 °C, no bias	1000 h	No functional fail
Temperature cycling	−40 to +125 °C, 500 cycles	—	No delamination
HAST	130 °C / 85% RH	96 h	No corrosion
ESD (HBM)	2 kV	—	Standard
ESD (CDM)	500 V	—	Standard
Latch-up	100 mA at 85 °C	—	Standard
Read disturb	85 °C, 10^{11} reads at max trim	—	No state change (margin 14.2× at 85 °C, max trim)

Test	Condition	Duration	Pass criterion
Write disturb	85 °C, adjacent cell, 10^9 writes	—	No state change (inhibit field 47.9% of E_c)

HTOL at 1 Hz. The HTOL cycling rate is 1 Hz over 1000 hours, giving 3.6×10^6 cycles — 0.15% of the design endurance of 2.45×10^9 cycles. This is the standard HTOL protocol for high-endurance non-volatile memory. Running at 1 MHz would consume 1,470× the endurance and would fail every device.

VII.8 Manufacturing Validation Plan

Phase	Duration	Content
Phase 1: Test structures	6 months	Fabricate HZO capacitors; P–V loop, coercive field, remanent polarization; above-pillar test structure
Phase 2: Single cells (Gate 2)	6 months	Fabricate individual memory cells; verify write/read; M1–M7 measurements
Phase 3: Small arrays (Gate 3)	12 months	Fabricate 1M-cell arrays; array operation; cell-to-cell variation; full-array functionality
Phase 4: Full chips	18 months	Fabricate full chips; yield; performance; power; thermal; reliability

Total: ~42 months from program start to production readiness.

VII.9 Complete Verification Checklist

1. Ferroelectric film thickness: ±1% (ellipsometry)
2. Hf:Zr ratio: ±2% (XPS)
3. Dielectric cap thickness: 7 nm ± 1 nm (ellipsometry)
4. CMP planarity: < 0.5 nm (AFM + SEM)
5. Ferroelectric phase: 50–70% orthorhombic (GI-XRD)
6. Remanent polarization: > 10 μC/cm² (P–V loop)
7. Coercive field: within 50% of target (P–V loop)
8. Pillar diameter: 20 nm ± 2 nm (SEM)
9. Pillar etch-back anisotropy: ≥ 3:1 (TEM cross-section)
10. TiN fill conformality: void-free (TEM cross-section)
11. Disk diameter: 40 nm ± 5% (SEM)
12. Disk sidewall roughness: < 1.5 nm RMS (AFM)
13. Single-cell read: within a factor of 3 of calculated resistance
14. Core conductivity: ≥ 0.1× design current (M1 ship gate)
15. Core radius: within 30% of 2.317 nm (M1 ship gate)
16. Above-pillar current: < 1 pA
17. ON/OFF ratio: > 10^5

18. Switching distribution: $\sigma \leq 0.20$
19. Loop area: $2.0–3.5 \times (P_s \cdot E_c)$
20. Retention: > 10 years (extrapolated)
21. Extended endurance (4-attempt cells): consistent with 2.45e9 spec
22. Array yield: > 80%
23. Cell variation: < 20%
24. Power: meets specification
25. Thermal: within operating range
26. Full chip: all functions work

If all 26 items pass, the device is ready for production.

Part VIII — Performance and Comparison

VIII.1 Overview

This Part provides the complete performance analysis and the comparison with existing memory technologies. All values reflect the final design including the dielectric cap, the self-aligned pillar-recess geometry, and the cold-corner behavior at 10 °C.

VIII.2 Density

Parameter	Value
Bitcell pitch	80 nm × 80 nm
Raw bitcell density	1.95 GB/cm²
Array efficiency	75%
Effective density	1.46 GB/cm²
Die area	100 mm²
User capacity	1.23 GB per die

VIII.3 Read Latency

Corner	On-chip (serial, first-access)	On-chip (pipelined steady-state)	System (add 3–5 ns)
TT (nominal)	4.71 ns	2.63 ns	~7.7–9.7 ns / ~5.6–7.6 ns
SS (worst)	4.88 ns	2.80 ns	~7.9–9.9 ns / ~5.8–7.8 ns

Serial vs pipelined. "Serial" (first-access) latency is the total time from row activation to data at the pad, summing all seven pipeline stages. "Pipelined" (steady-state) latency is the time from column-address-select to data, when the row is already open and bank/row/sub-array decode overlap with the previous access. Memory datasheets typically quote both, with tRCD (serial) and tCL (pipelined) as separate parameters. The system-level read latency (~7–10 ns serial, ~5.6–7.8 ns pipelined) includes controller overhead.

Baselines: DDR5-8000 system read latency ~55–70 ns; 3D NAND read latency ~50 µs (system).

VIII.4 Read Bandwidth

Parameter	Value
Peak bandwidth (64-bit, 6.4 Gbps)	51.2 GB/s
Sustained read throughput	51.2 GB/s (bus-limited)

VIII.5 Write Latency

Condition	Per word	System (add 3–5 ns)
25 °C, nominal VDD	45.2 ns avg / 49.5 ns max	~48 / 52 ns
85 °C, nominal VDD	45.2 ns avg / 49.5 ns max	~48 / 52 ns
10 °C, −5% VDD (cold corner)	49.5 ns (all 4 attempts)	~52 ns

Comparison to DDR5-8000: system write latency is 50–65 ns. The Vortex FeRAM is **comparable** to DDR5 (within 10–20%).

Comparison to NAND SLC: system write latency is ~100 µs. The Vortex FeRAM is **~2,000× faster**.

VIII.6 Write Bandwidth

Condition	Bandwidth
Pure-write (4 banks active)	0.7 GB/s
50/50 mixed read/write	0.33 GB/s
90/10 read-heavy	0.6 GB/s

The write bandwidth is 0.7 GB/s, not 51.2 GB/s. The write-verify architecture fundamentally limits write throughput: each word takes 45.2 ns to complete, so the maximum sustained write rate is 1/(45.2 ns) × 8 bytes = 0.177 GB/s per bank. Four banks running in parallel give 0.7 GB/s.

This is the fundamental trade-off of write-verify memory: **write throughput is 20–70× lower than read throughput**. The Vortex FeRAM is a read-fast, write-slow memory. Applications that need write throughput at DDR5-class rates should use DDR5; applications that benefit from fast reads and non-volatility should use this product.

VIII.7 Write Energy

Parameter	Value	Comparison
Cell write energy (avg, 25 °C)	0.88 fJ	11,364× lower than DDR5 (cell); 568× lower than NAND (cell)
Cell write energy (cold corner)	0.94 fJ	—

Parameter	Value	Comparison
System write energy (including I/O)	~6.5 pJ per bit	Comparable to DDR5 (~10 pJ)

Cell vs system energy. The 0.88 fJ figure is the cell-level write energy. At the system level, the I/O charge on the external bus (88 active pins × 4 pF × 1.8 V^2 × 0.5 activity) dominates at ~6.5 pJ per bit. This is comparable to DDR5's system write energy (~10 pJ per bit). The 11,364× advantage applies at the cell level only.

VIII.8 Power

State	Power	Per GB
Idle (Tj = 87 °C)	109 mW	66 mW/GB
50% duty (read-saturated)	1021 mW	621 mW/GB
100% duty (read-saturated)	1934 mW	1176 mW/GB
100% duty (write-saturated)	~221 mW	134 mW/GB

DDR5-8000 comparison: 250–375 mW/GB at idle. The Vortex FeRAM is **4–6× lower** at idle (66 vs 250–375).

At 50% duty: the Vortex FeRAM is **1.7–2.5× higher** than DDR5 (621 vs 250–375 mW/GB). The Vortex FeRAM is not a power-efficiency tier at sustained operation.

At 100% duty (read-saturated): the Vortex FeRAM requires thermal pad (θ_{JA} = 15 °C/W) or ambient ≤80 °C with the standard package (θ_{JA} = 20 °C/W).

VIII.9 Endurance

Parameter	Value
Endurance at 1% fail (Tj = 87.2 °C)	2.45×10^9 cycles
Endurance at 0.1% fail	7.7×10^8 cycles
vs NAND SLC	24,500× higher (2.45e9 vs 1e5)
vs DDR5	4×10^6× lower (2.45e9 vs 1e16)

VIII.10 Cost per User GB

Parameter	Value
Die cost (all-in, 100 mm^2)	$6.85
User capacity	1.23 GB

Parameter	Value
Cost/GB primary (wafer $2,500)	$5.54
Cost/GB volume (wafer $1,200)	$3.34

VIII.11 Comparison with DDR5-8000

Parameter	DDR5-8000	Vortex FeRAM
Density (GB/cm², die)	2.7–3.5	1.46 (0.42–0.54×)
Read latency (system)	55–70 ns	~7.7–9.7 ns (6–9× faster)
Write latency (system)	50–65 ns	~48–52 ns (comparable)
Read bandwidth	~51 GB/s	51.2 GB/s (comparable)
Write bandwidth	~51 GB/s	0.7 GB/s (73× lower)
Write energy per cell	~10 pJ	0.88 fJ (11,364× lower)
Endurance	10^{16}	2.45×10^{9} (4×10^{6}× lower)
Non-volatile	No	Yes
Power/GB (idle)	250–375 mW/GB	66 mW/GB (4–6× lower)
Power/GB (50% duty)	250–375 mW/GB	621 mW/GB (1.7–2.5× higher)
Cost/GB (primary)	$5–8	$5.54 (comparable)

VIII.12 Comparison with 3D NAND SLC

Parameter	3D NAND SLC	Vortex FeRAM
Density (GB/cm², die)	23–30	1.46 (0.05–0.06×)
Read latency (system)	~50 µs	~7.7–9.7 ns (5,500–7,100× faster)
Write latency (system)	~100 µs	~48–52 ns (2,000× faster)
Write energy per cell	~500 fJ	0.88 fJ (568× lower)
Endurance	10^{5}	2.45×10^{9} (24,500× higher)
Non-volatile	Yes	Yes
Cost/GB (primary)	$0.50–1.00	$5.54 (5–11× more expensive)

VIII.13 Comparison with MRAM STT

Parameter	MRAM STT	Vortex FeRAM

Parameter	MRAM STT	Vortex FeRAM
Density (GB/cm², die)	1.5–2.0	1.46 (0.7–1.0×)
Read latency (system)	~20–30 ns	~7.7–9.7 ns (2.5–3.3× faster)
Write latency (system)	~20–30 ns	~48–52 ns (1.7–2.5× slower)
Write energy per cell	~100 fJ	0.88 fJ (114× lower)
Endurance	$\sim 10^{12}$	2.45×10^9 (400× lower)
Cost/GB (primary)	$3–5	$5.54 (comparable to slightly higher)

VIII.14 Complete Technology Comparison

Metric	DDR5-8000	3D NAND SLC	MRAM STT	Vortex FeRAM
Density (GB/cm², die)	2.7–3.5	23–30	1.5–2.0	1.46
Read latency (system)	55–70 ns	~50 µs	~20–30 ns	~7.7–9.7 ns
Write latency (system)	50–65 ns	~100 µs	~20–30 ns	~48–52 ns
Read bandwidth	~51 GB/s	1–2 GB/s	1–3 GB/s	51.2 GB/s
Write bandwidth	~51 GB/s	1–2 GB/s	1–3 GB/s	0.7 GB/s
Write energy per cell	~10 pJ	~500 fJ	~100 fJ	0.88 fJ
Endurance	10^{16}	10^5	10^{12}	2.45×10^9
Non-volatile	No	Yes	Yes	Yes
Cost/GB (primary)	$5–8	$0.50–1.00	$3–5	$5.54

VIII.15 Positioning

VIII.15.1 What works well

- **Read latency:** 6–9× faster than DDR5, 5,500–7,100× faster than NAND, 2.5–3.3× faster than MRAM.
- **Read bandwidth:** 51.2 GB/s (DDR5-class, much faster than NAND or MRAM).
- **Write latency:** Comparable to DDR5 (within 10–20%), 2,000× faster than NAND.
- **Cell write energy:** 0.88 fJ, 11,364× lower than DDR5, 568× lower than NAND, 114× lower than MRAM.
- **Non-volatility:** >10 years at 85 °C. Instant power-off. No refresh.
- **Idle power:** 66 mW/GB, 4–6× lower than DDR5.
- **Endurance:** 2.45×10^9 cycles, 24,500× better than NAND SLC.
- **Cost/GB:** Comparable to DDR5 ($5.54 vs $5–8 primary).

VIII.15.2 What is weak

- **Write bandwidth:** 0.7 GB/s, 73× lower than DDR5. The write-verify architecture fundamentally limits write throughput. This is not a design flaw — it is an inherent property of write-verify memory.
- **Density:** 1.46 GB/cm², comparable to MRAM, but much lower than NAND (23–30 GB/cm²) or DDR5 (2.7–3.5 GB/cm²). The single-layer design and 80 nm pitch limit density.
- **Endurance vs DRAM:** 2.45×10^9 cycles, 4×10^6× lower than DRAM. Not a DRAM endurance replacement.
- **Power at 50%+ duty:** 621 mW/GB, 1.7–2.5× higher than DDR5. The Vortex FeRAM is not a power-efficiency tier at sustained operation.
- **Cost vs NAND:** $5.54/GB, 5–11× more expensive than NAND SLC. Not positioned as a NAND price competitor.
- **Operating temperature:** Writes require ≥10 °C; below that, cold-start warm-up or extended ladder. Reads are functional across −40 to 125 °C.

VIII.15.3 Bottom line

The Vortex FeRAM is a **read-fast, non-volatile, endurance-focused** memory. It is positioned for applications where:

- Non-volatility is required (data survives power-off).
- Read latency is the dominant performance requirement.
- Endurance of 2.45×10^9 cycles is sufficient (embedded, industrial, automotive, hot-storage).
- Write throughput of 0.7 GB/s is acceptable (write-light workloads).
- Cost per GB comparable to DDR5 is acceptable (not chasing NAND cost).

It is not a DDR5 replacement, not a NAND replacement, and not a DRAM endurance replacement. It occupies a distinct position: **read-fast, non-volatile, endurance-differentiated, NAND-class write bandwidth**.

Part IX — Interface Specification

IX.1 Interface Overview

The memory uses a synchronous custom DDR-style interface. It is **not** a JEDEC-compliant DDR5 interface. The electrical characteristics are custom to this device; the command protocol is device-specific. A custom memory controller (or a bridge chip) is required; the device cannot be connected directly to a DDR5 socket or controller.

IX.1.1 Interface characteristics

Parameter	Value
Data width (external)	64-bit (DQ[0:63])
ECC	8-bit SECDED internal, transparent to host
Command/address width	32 pins (CA[0:31]), 4-cycle command word = 256 bits
Data rate (per pin)	6.4 Gbps
Peak bandwidth (64-bit)	51.2 GB/s
DQS signaling	Single-ended
Command clock	DDR at 3.2 GHz (1/2 data rate)
Package	BGA-324
Burst length	BL16 (16 beats × 64 bits = 1,024 bits per burst)
Address bits	34

IX.1.2 Voltage rails

Rail	Voltage	Purpose
VDD_CORE	1.0 V ± 5%	Digital logic, comparator
VDD_PLL	1.0 V ± 5%	Clock generation
VDDQ	1.8 V ± 5%	Standard I/O (DQ, DQS, CA)
VDD_ANALOG	1.8 V ± 5%	DAC, ADC, TIA, reference
VPP (inhibit)	2.5 V ± 5%	Inhibit driver
VPROG (write-boost)	3.5 V	Write driver, from on-die charge pump
V_WL (word-line boost)	1.5 V	SWL drive, from on-die charge pump

Interface standard note. VDDQ = 1.8 V matches DDR3L, not DDR5 (which uses 1.1 V). The 1/2 data-rate command bus is DDR-style, not QDR. The device uses a custom protocol

throughout. It is not compatible with any JEDEC DDR5 socket or controller without a custom bridge.

IX.2 Command Set

Opcode	Mnemonic	Description
0x00	NOP	No operation
0x01	ACT	Activate a row in a bank
0x02	PRE	Precharge a bank
0x03	PRE_ALL	Precharge all banks
0x04	RD	Read
0x05	WR	Write
0x06	WRV	Write-verify (write + read-verify in hardware)
0x07	ERS	Erase (write n=0)
0x08	REF	Refresh reference cells
0x09	MRS	Mode register set
0x0A	MRR	Mode register read
0x0B	MPC	Multi-purpose command (DQS training, duty-cycle adjust)
0x0C	TRIM	Read-voltage trim set
0x0D	CAL	Calibration
0x0E	STAG	Status read
0x0F	ECCW	ECC write (write check bits explicitly)
0x10	ECCR	ECC read (read check bits without correction)
0x11	PWRD	Power-down enter
0x12	SWAP	Polarity swap (polarity inversion for imprint mitigation)

19 opcodes total. Bits 0x13–0x1F reserved. Bit 0x20–0x3F reserved for future expansion. The upper opcode bits (0x40–0xFF) are reserved for vendor-specific extensions.

IX.3 Pin-Out — BGA-324

The package uses an 18 × 18 BGA grid (324 balls). Balls are numbered A1–A18, B1–B18, …, R1–R18. The letters I and O are used (18 letters total, A through R); this is documented on the package datasheet.

Ball	Signal	Function	Voltage

Ball	Signal	Function	Voltage
A1–A4	VDD	1.0 V core supply	1.0 V
A5–A8	VSSQ	Ground (I/O domain)	0 V
A9–A12	VDDQ	1.8 V I/O supply	1.8 V
A13–A16	VSS	Ground (core domain)	0 V
A17–A18	VPP	2.5 V inhibit supply	2.5 V
B1–B16	DQ[0:15]	Data bus, byte lanes 0 and 1	1.8 V
B17	DQS[0]	Data strobe, byte lanes 0–1 (single-ended)	1.8 V
B18	DQS[1]	Data strobe, byte lanes 2–3 (single-ended)	1.8 V
C1–C16	DQ[16:31]	Data bus, byte lanes 2 and 3	1.8 V
C17	DQS[2]	Data strobe, byte lanes 4–5	1.8 V
C18	DQS[3]	Data strobe, byte lanes 6–7	1.8 V
D1–D16	DQ[32:47]	Data bus, byte lanes 4 and 5	1.8 V
D17	DQS[4]	Data strobe, byte lanes 8–9	1.8 V
D18	DQS[5]	Data strobe, byte lanes 10–11	1.8 V
E1–E16	DQ[48:63]	Data bus, byte lanes 6 and 7	1.8 V
E17	DQS[6]	Data strobe, byte lanes 12–13	1.8 V
E18	DQS[7]	Data strobe, byte lanes 14–15	1.8 V
F1–F16	CA[0:15]	Command/address, lower 16 bits	1.8 V
F17	CS_N	Chip select (active low)	1.8 V
F18	CKE	Clock enable	1.8 V
G1	CLK	Differential clock P	1.8 V
G2	CLK_N	Differential clock N	1.8 V
G3	DQS_C	DQS reference clock P	1.8 V
G4	DQS_C_N	DQS reference clock N	1.8 V
G5	ODT	On-die termination control	1.8 V
G6	RESET_N	Reset (active low)	1.8 V
G7	ALERT_N	Error alert (active low)	1.8 V
G8	TEN	Test enable (factory)	1.8 V
G9–G12	VPROG	Write-boost supply (3.5 V)	VPROG

Ball	Signal	Function	Voltage
G13–G17	JTAG	TCK, TMS, TDI, TDO, TRST_N	1.8 V
G18	VPROG_SENSE	Kelvin sense for VPROG regulation	VPROG
H1–H16	CA[16:31]	Command/address, upper 16 bits	1.8 V
H17	PARITY_CA	Command/address parity bit	1.8 V
H18	VREF_CA	Reference voltage for CA receiver (external optional)	0.6 V nominal
J1–J8	VREF_DQ[0:7]	Reference voltage for DQ byte lanes 0–7	0.6 V nominal
J9–J18	VDD / VSS / VDDQ / VSSQ	Power and ground distribution	Various
K1–R17	VDD / VSS / VDDQ / VSSQ	Power distribution; remaining balls for future expansion	Various
R18	V_WL_SENSE	Kelvin sense for V_WL regulation	V_WL

IX.4 Pipeline Timing

Parameter	Value	Description
t_{RCD}	1.6 ns	Row active to CAS
t_{CL}	2.18 ns	CAS latency
t_{Burst}	2.5 ns	Burst length BL16 at data rate
t_{RAS}	6.5 ns	Active to precharge (minimum)
t_{RP}	1.0 ns	Precharge
t_{RC}	7.5 ns	Row cycle time
t_{RTP}	4 cycles	Read to precharge
t_{WR}	22 ns	Write recovery
t_{WL}	4 cycles	Write latency
t_{WTR}	24 ns	Write to read

Serial vs pipelined latency. The on-chip serial first-access read latency is 4.71 ns at TT (row activation through data at the pad, all seven pipeline stages). The on-chip pipelined steady-state read latency is 2.63 ns at TT (column-address-select to data, with the row already open and prior-cycle decode overlapping). The tRCD, tCL, and tBurst parameters above are the DDR-style timing reference for controller compatibility; the serial and pipelined on-chip figures are the direct timing numbers.

Burst length BL16 delivers 16 beats × 64 bits = 1,024 bits = 16 words per burst.

IX.5 On-Die Termination and DQS Training

IX.5.1 ODT values

ODT mode	Available values
Rtt_nom (used during read)	34, 40, 48, 60, 80, 120, 240 Ω
Rtt_wr (used during write)	40, 60, 80, 120, 240 Ω
Rtt_park (used when idle on active channel)	34, 40, 48, 60, 80, 120, 240 Ω
ODT high-Z (disabled)	Yes, via MRS

IX.5.2 DQS training protocol

1. **Write leveling:** Align DQS to CK at the memory. Sweeps DQS delay in 5 ps steps over ±500 ps.
2. **Read gate training:** Align read DQS to internal clock at the controller.
3. **Read data eye training:** Sweep DQS delay to center the read data eye.
4. **Write data eye training:** Sweep DQ delay to center the write data eye.
5. **Vref training:** Sweep Vref to center the data eye at the receiver.

Training time: ~2.8 ms total. Training is run at power-on and whenever the temperature changes by more than 10 °C.

IX.6 Power Sequencing

IX.6.1 Power-on sequence

1. VDD_CORE (1.0 V) first.
2. VDD_PLL (1.0 V) within 100 µs.
3. VDDQ (1.8 V), VDD_ANALOG (1.8 V), VPP (2.5 V) next, within 1 ms of VDD_CORE.
4. VPROG (3.5 V) after VDDQ has reached 90% of nominal.
5. V_WL (1.5 V) last.

IX.6.2 Power-down sequence

Reverse order: V_WL off first, then VPROG, then VPP, then VDDQ/VDD_ANALOG, then VDD_PLL, then VDD_CORE.

IX.6.3 Hot-plug rules

Power-on reset monitors all rails and disables VPROG output until VDD and VDDQ are both up. A hot-plug detection circuit pulls VPROG to 0 V during any rail transition where VDDQ is not within 90% of nominal. The device must not be hot-plugged with VPROG live and VDDQ off.

IX.7 Error Reporting

Error type	Reporting mechanism
Single-bit ECC correctable	ALERT_N pulse; status bit set; ECC auto-corrected
Double-bit ECC detected	ALERT_N pulse; status bit set; operation retried
Uncorrectable (triple+)	ALERT_N continuous; status bit set; block marked bad
Bad-block event	ALERT_N pulse; status bit set; bad-block list updated in OTP
Brown-out (VDD_CORE < 0.9 V)	RESET_N asserted; in-flight write aborted; ECC catches on next read
Over-temperature (Tj > 120 °C)	ALERT_N pulse; status bit set; write rate throttled

IX.8 Read-Modify-Write for Partial-Word Writes

The memory supports full 64-bit word writes without read-modify-write. Partial-word writes (e.g., 32-bit half-word) trigger the read-modify-write sequence:

1. Read the 72-bit ECC-protected word.
2. Modify the affected bytes.
3. Recompute the 8-bit ECC check bits.
4. Write back the full 72 bits.

RMW cost: 4.71 ns (read serial) + 45.2 ns (write avg) = **~50 ns**. The device includes a 10 KB per-bank write-combining buffer that merges partial writes before commit; typical workloads see 80%–90% reduction in RMW events.

IX.9 Complete Interface Summary

Parameter	Value
Data width	64-bit + 8-bit ECC (72/64)
Command/address	32 pins, 4-cycle command word (256 bits)
Data rate	6.4 Gbps
Peak bandwidth	51.2 GB/s
Burst length	BL16 (16 beats × 64 bits)
VDD rails	1.0 V core, 1.0 V PLL, 1.8 V I/O, 1.8 V analog, 2.5 V inhibit, 3.5 V VPROG, 1.5 V V_WL

Parameter	Value
Command set	19 opcodes
Package	BGA-324
DQS	Single-ended
JTAG	IEEE 1149.1, 5 pins
Boot time	~3 ms (including DQS training)
Recalibration trigger	Hourly or ±5 °C temperature change

Part X — Health and Safety

X.1 Production Hazards

The production of the memory uses two ALD precursors that are corrosive and require inert-atmosphere handling. These are not common knowledge for a general engineering reader, so the hazards are noted here.

X.1.1 Hafnium chloride ($HfCl_4$)

- White crystalline solid, sublimes at 317 °C.
- Reacts with water to form HCl (hydrochloric acid).
- Corrosive to skin, eyes, and the respiratory tract.
- **Handling:** inert atmosphere (N_2 or Ar), dry conditions. Standard ALD precursor handling procedure.
- **Exhaust:** HCl scrubber required.

X.1.2 Zirconium chloride ($ZrCl_4$)

- White crystalline solid, sublimes at 331 °C.
- Reacts with water to form HCl.
- Corrosive, same hazards as $HfCl_4$.
- **Handling:** same as $HfCl_4$.

X.1.3 The HCl byproduct

The ALD reactions produce HCl (4 moles per mole of HfO_2 or ZrO_2). The exhaust must be scrubbed with a wet scrubber or thermal abatement. Standard semiconductor fab practice.

X.1.4 CMP slurry

The CMP planarization steps (pillar planarization and cap planarization) use silica-based slurry at pH 10–11. Standard CMP slurry handling applies: eye protection, gloves, and slurry waste treatment. Silica particles are a respiratory irritant if dried and aerosolized; slurry must be kept wet until waste treatment. Standard CMP tooling includes the required safety controls.

X.1.5 Standard semiconductor hazards

Standard semiconductor fabrication hazards (silane, hydrofluoric acid, chlorine etchants, various solvents) apply throughout the flow. These are common to all CMOS fabs and are handled by the foundry's standard safety infrastructure. No additional hazards are introduced by the HZO ferroelectric process beyond the two Hf/Zr precursors and the HCl byproduct.

X.2 Post-Production Safety

The finished memory chips are standard CMOS devices with no hazardous materials exposed. HZO is a stable ceramic; no special handling is required for the finished chip. The finished chip is safe for consumer, commercial, and industrial use.

X.3 Handling of the Finished Memory

The finished memory is a BGA-324 device, handled like any other BGA memory device:

- **ESD protection:** handle in an ESD-controlled environment (wrist strap, ESD mat).
- **Moisture sensitivity:** MSL 3 (168 hours floor life at 30 °C, 60% RH). Bake at 125 °C for 24 hours if floor life is exceeded.
- **Reflow:** standard J-STD-020 Pb-free reflow, peak 260 °C.
- **Storage:** dry pack with desiccant, ≤ 40 °C, ≤ 90% RH.

X.4 Environmental Compliance

The memory is compliant with:

- **RoHS** (Restriction of Hazardous Substances) — no lead, cadmium, mercury, hexavalent chromium, PBB, or PBDE.
- **REACH** (Registration, Evaluation, Authorization, and Restriction of Chemicals) — no SVHC above threshold.
- **Halogen-free** (per IEC 61249-2-21).

Part XI — Supply Chain and Tool Requirements

XI.1 Reference to Part IV

The equipment list, foundry requirements, and supply chain are documented in Part IV.11 (Equipment List), Part IV.12 (Foundry Requirements), and Part IV.13 (Supply Chain). This Part provides the consolidated supplier information and the dual-source strategy that complements those sections.

XI.2 Foundry Strategy (Dual-Source)

The Vortex FeRAM is designed for dual-source manufacturing. Two qualified foundries are identified:

Foundry	Node	Status
TSMC 28HPC+	28 nm CMOS with $CoSi_2$ S/D	Primary qualified
GlobalFoundries 22FDX	22 nm FD-SOI (compatible process options)	Secondary qualified

Both foundries offer the required $CoSi_2$ source/drain, 10 V extended-drain transistors, and HZO-capable ALD. If one becomes unavailable, the other can supply. This dual-source strategy eliminates single-foundry risk for this product.

XI.3 Precursor Supplier Categories

All precursors are available from multiple suppliers. No single-supplier risk exists for any material.

Material	Supplier category	Lead time
$HfCl_4$ (99.99%)	ALD precursor supplier (multiple)	6–10 weeks
$ZrCl_4$ (99.99%)	ALD precursor supplier (multiple)	6–10 weeks
$TiCl_4$ (99.99%)	Standard semiconductor precursor	4–8 weeks
$Al(CH_3)_3$ (TMA)	Standard semiconductor precursor	4–8 weeks
NH_3 (99.999%)	Standard semiconductor gas	2–4 weeks
H_2O (18.2 MΩ·cm)	In-house ultrapure water system	Continuous
CMP slurry (silica)	Standard CMP supplier (multiple)	4–6 weeks
300 mm Si wafers	Standard wafer supplier (multiple)	4–8 weeks
BGA-324 substrate	Package supplier (multiple)	8–12 weeks
Gold bond wire (25 µm)	Standard	2–4 weeks

Material	Supplier category	Lead time
Die attach epoxy	Standard	2–4 weeks

XI.4 Tool Availability

All required tools are standard semiconductor equipment, available from multiple vendors:

Tool class	Vendor category	Dual-source?
ALD (TiN, Al_2O_3, HfO_2, ZrO_2)	Applied Materials, ASM, Lam Research	Yes
CMP with in-situ endpoint	Applied Materials (Reflexion LK), Ebara (F-REX)	Yes
RTA furnace	Applied Materials, ULVAC, Annealsys	Yes
193 nm immersion lithography	ASML (NXT series)	Single-source (industry standard)
RIE etcher (Cl_2/Ar, Cl_2/BCl_3)	Lam Research, Applied Materials, TEL	Yes
PECVD	Applied Materials, Lam Research	Yes
PVD (Cu seed)	Applied Materials, Evatec	Yes
Electroplating	Applied Materials, Lam Research	Yes

Only immersion lithography is single-source (ASML is the global monopoly), but this is industry-standard for all sub-40 nm nodes and does not constitute a product-specific risk.

XI.5 Qualification Requirements

Before production, the following must be qualified with the foundry:

1. **$CoSi_2$ source/drain process module:** confirm stable to 700 °C with the foundry's process flow.
2. **HZO ferroelectric ALD:** confirm the foundry can deposit HfO_2/ZrO_2 with the required 50:50 ratio and 15.60 nm thickness.
3. **RTA at 450 °C:** confirm the foundry's thermal budget can accommodate the ferroelectric anneal (this is the highest-temperature step in the ferroelectric back-end).
4. **CMP with in-situ interferometry endpoint:** confirm the foundry's CMP tool has the required in-situ endpoint detection for the pillar and cap planarization steps.
5. **193 nm immersion + LELE for 20 nm pillar CD:** confirm the foundry can achieve the required CD control at 20 nm.
6. **HCl scrubber capability:** confirm the foundry has the exhaust abatement required for the $HfCl_4$ and $ZrCl_4$ ALD reactions.

All six are standard 28 nm CMOS fab capabilities. The $CoSi_2$ module and HZO ferroelectric capability are the two non-standard items and must be confirmed at foundry selection.

Part XII — Software Driver and Usage Guidelines

XII.1 Purpose

This Part specifies the software interface (driver API) and the operational guidelines that any host system must follow to achieve the endurance and lifetime characteristics stated in Parts I and VIII. Because the Vortex FeRAM has very high cycle endurance (2.45×10^9 cycles), wear leveling is optional for most workloads, but recommended for workloads with known asymmetric patterns.

XII.2 Lifetime vs Sustained Write Rate

The Vortex FeRAM has 2.45×10^9 cycles at 1% fail. For a 1.23 GB die with a typical write amplification factor of 1.0, the lifetime under continuous writing depends on the sustained write rate:

Sustained write rate (per die)	Expected lifetime (1% fail)	Recommended use
≤ 100 MB/s	≥ 100 years	Any workload
≤ 500 MB/s	≥ 20 years	Long-lived storage, embedded, industrial
≤ 0.7 GB/s (peak write BW)	≥ 14 years	Write-heavy workloads with wear leveling

The Vortex FeRAM does not need wear leveling for most workloads. With wear leveling (DWLV, §XII.4), the effective endurance is extended by the wear-leveling factor (typically 5–10×), giving effective lifetime of 70–140 years at peak write rate.

XII.3 Burst-Write Buffer

If an application bursts at higher rates but sustains at a lower rate, the burst must be smoothed before it reaches the memory. Buffer requirements (drained at 100 MB/s):

Burst duration at 1 GB/s	Buffer size
1 ms	0.9 MB
10 ms	9 MB
100 ms	90 MB
1 second	900 MB

The buffer should be implemented in host DRAM (not in the FeRAM itself), because buffering

in the FeRAM converts the buffer writes into FeRAM writes.

XII.4 Wear Leveling (Optional)

XII.4.1 The algorithm

The recommended wear-leveling algorithm (when enabled) is **DWLV hybrid** (Dynamic Wear Leveling with Variable window), an 8-bit counter per 4 KB block:

1. Each 4 KB block has an 8-bit cycle counter, stored in a reserved metadata area (0.3 MB for a 1.23 GB die, 0.025% of user capacity).
2. New writes go to the block with the lowest counter.
3. Blocks with counter ≥ 128 (2× mean) are frozen (allowed to cool).
4. When a block's counter reaches 200, its data is migrated to a fresh block; the block is added to the spare pool.
5. Overshoot factor: ≤ 5× (realistic).

Wear leveling is optional. The Vortex FeRAM's 2.45×10^9 cycle endurance exceeds any practical workload lifetime for most applications. Wear leveling is recommended only for workloads with known asymmetric patterns or for very long deployment lifetimes.

XII.4.2 Metadata storage

Parameter	Value
Metadata per 4 KB block	1 byte (8-bit counter)
Metadata for 1.23 GB die	0.3 MB (0.025% of user capacity)
Location	Reserved area, ECC-protected
Write amplification (DWLV overhead)	1.1–1.2×
Overshoot factor	≤ 5×

XII.5 SWAP Command (Polarity Swap, Optional)

XII.5.1 What SWAP does

The memory cell encoding is: n=1 for "written one state," n=0 for "written zero state." If a block sees asymmetric cycling (e.g., 80% of writes are to n=1, 20% to n=0), the n=1 cells fatigue faster than the n=0 cells. Over time, this creates imprint: the n=1 cells have lower remanent polarization than the n=0 cells, degrading the read margin for the n=1 state.

The SWAP command inverts the encoding for a block: n=1 becomes "written zero state," n=0 becomes "written one state." After SWAP, the previously over-cycled n=1 cells are now written as n=0 and vice versa. This equalizes the cycling across the two polarization states.

XII.5.2 SWAP protocol

1. Each 4 KB block has a 32-bit write counter.
2. When the counter reaches 10^5, the controller issues a SWAP command to the block.
3. The memory reads the block, inverts the polarity encoding, writes back, and resets the block's 1-bit polarity flag.
4. The counter resets to zero.

SWAP is optional. It is recommended for workloads with asymmetric read/write patterns. SWAP storage cost (when enabled): 1 bit per 64-bit word = 1.56% storage overhead.

XII.6 Address Scrambling

The memory controller must implement address scrambling to prevent hot-spot wear. Without scrambling, sequential writes hit the same physical addresses, causing concentration of wear at low addresses.

Parameter	Value
Scramble formula	Physical address = logical address XOR scramble_key
Scramble key	32-bit, derived from die serial number, stored in OTP
Hardware cost	1 XOR per address (negligible area)
Software transparency	Fully transparent to driver

XII.7 Read-Modify-Write and Write-Combining Buffer

Partial-word writes (less than 64 bits) trigger RMW. Cost:

- Read the 72-bit ECC-protected word: 4.71 ns (serial).
- Modify affected bytes.
- Recompute ECC.
- Write back the full 72 bits: 45.2 ns (average).
- **Total RMW: ~50 ns.**

The device includes a 10 KB per-bank write-combining buffer (160 KB total) that merges partial writes before commit. The buffer reduces RMW events by 80%–90% for typical workloads. The driver should flush the buffer before power-down or before any read to a partially written word.

XII.8 Reference Cell Refresh

The reference cells (1,024 per bank in a separate column) are the median-current reference for the sense amplifier. Two refresh mechanisms:

- **Timer-based:** refresh every 10^5 writes to that bank.

- **Error-rate-triggered:** when the single-bit ECC error rate per bank rises above 2× baseline, issue a refresh.

Parameter	Value
Reference cells per bank	1,024 (dedicated column)
Refresh frequency	Every 10^5 writes to that bank, or on ECC-trigger
Refresh cost	1% of write bandwidth
Refresh operation	Re-write reference cell to n=1
Scheduling	Background; not on critical path

XII.9 Driver API

Function	Signature	Purpose
memory_init	`int memory_init(struct config *cfg)`	Boot: reset, load trims, DQS training, warm-up
memory_read	`int memory_read(uint64_t addr, size_t len, void *buf)`	Read data; ECC auto-checked and corrected
memory_write	`int memory_write(uint64_t addr, size_t len, const void *buf)`	Write with verify; RMW for partial words
memory_erase	`int memory_erase(uint64_t block_addr)`	Erase block to all-zeros
memory_format	`int memory_format(struct format_params *p)`	First-time device setup (bad-block scan, metadata initialization)
memory_recalibrate	`int memory_recalibrate(void)`	Force DQS/trim recalibration (for temperature events)
memory_swap	`int memory_swap(uint64_t block_addr)`	Polarity swap on block (optional)
memory_swap_exclusive	`int memory_swap_exclusive(uint64_t block_addr)`	SWAP with block lock (blocks all other accesses during swap)
memory_trim_set	`int memory_trim_set(uint8_t v_read_trim, uint8_t i_ref_trim)`	Set read trim (both 4-bit fields)
memory_set_ecc	`int memory_set_ecc(enum ecc_mode mode)`	Configure ECC mode (internal SECDED or external)
memory_get_capacity	`struct capacity memory_get_capacity(void)`	Report raw vs user capacity

Function	Signature	Purpose
memory_status	`struct status memory_status(void)`	Read status register, error counters, temperature
memory_sleep	`int memory_sleep(enum power_state state)`	Enter P1/P2/P3
memory_wake	`int memory_wake(void)`	Exit sleep; waits 50 µs if from P3
memory_scrub	`int memory_scrub(uint64_t addr, size_t len)`	Refresh ECC state; correct single-bit errors proactively
memory_get_wear	`struct wear_info memory_get_wear(uint64_t addr)`	Report block wear metrics

XII.9.1 Error codes

Code	Meaning	Recommended handling
0	Success	—
-EIO (single-bit)	Single-bit ECC corrected	Log, continue
-EIO (double-bit)	Double-bit ECC detected	Retry once; if retry succeeds, log and continue; if fails, mark block bad
-EIO (uncorrectable)	Uncorrectable error	Mark block bad; host notified
-ENOMEM	Insufficient buffer	Caller provides larger buffer
-EBUSY	Device busy (in calibration or training)	Wait and retry
-EINVAL	Invalid argument	Check argument
-EALIGN	Address not aligned to 64-bit word	Realign address
-ETEMP	Over-temperature (Tj > 120 °C)	Throttle writes (driver handles automatically)
-EBROWN	Brown-out detected during operation	Retry operation
-ENOSPC	No spare blocks available (bad-block reserve exhausted)	Device end-of-life; format required

XII.9.2 Thread safety

All driver functions are thread-safe. The driver uses an internal mutex to serialize access to the memory device. Multi-threaded callers can issue read/write calls concurrently without additional locking. Callers that issue a SWAP command must use `memory_swap_exclusive` to block other accesses during the swap.

XII.9.3 Timing guarantees

- **memory_read:** returns within 4.88 ns (SS corner, serial) + DMA time.
- **memory_write:** returns within 49.5 ns (max) + DMA time.
- **memory_swap:** returns within ~10 µs per 4 KB block (including read + invert + write + verify).
- **memory_wake from P3:** returns within 50 µs + 10 ns.

XII.10 Driver-Level Recommendations

- **Enable address scrambling** at mount time (all deployments).
- **Enable the write-combining buffer** for workloads with partial-word writes.
- **Schedule reference refresh** in the background every 10^5 writes per bank, or use ECC-error-triggered refresh.
- **Monitor temperature** and throttle writes if Tj > 120 °C (driver handles automatically).
- **Enable DWLV wear leveling** only for workloads with known asymmetric patterns or very long deployment lifetimes (optional).
- **Schedule SWAP** for workloads with asymmetric cycling (optional).
- **Log ECC error rates** via the status register; a rising single-bit rate indicates the block is approaching its wear limit.
- **Handle double-bit errors** by retrying once before marking the block bad.

Part XIII — Cost Analysis

XIII.1 Overview

This Part provides the cost analysis of the Vortex FeRAM. All figures are 2026 estimates in US dollars. They exclude non-recurring engineering costs (mask NRE, qualification costs, IP licensing), which are program-level and not part of the product's per-unit economics.

Two pricing bases are given:

- **Primary** — $2,500/wafer for 28 nm LELE. Conservative; usable for planning.
- **Volume** — $1,200/wafer. Target-volume foundry pricing; achievable only at high order volume.

XIII.2 Die Cost Breakdown

Cost component	Value
Wafer processing per good die (primary)	$5.23
Wafer probe test	$0.17
Packaged test (including wake-up cycle)	$0.30
Burn-in	$0.05
ESD protection (VPROG clamp)	$0.10
BGA-324 package	$1.00
Total die cost (primary)	**$6.85**

Wafer processing per good die is calculated as (wafer cost) / (615 gross dies per wafer × yield). With 77.7% yield: $2,500 / (615 × 0.777) = $5.23. Package cost of $1.00 is the target-volume price; at low volume add $1.00.

XIII.3 Cost per User GB

User capacity is the raw capacity after ECC (12.5%), bad-block reserve (2%), and — optionally — the SWAP polarity flag (1.56% when enabled). The Vortex FeRAM ships without SWAP enabled by default (SWAP is optional), so the user capacity uses 12.5% + 2% overhead only:

Parameter	Value
Effective density (75% array efficiency)	1.46 GB/cm²
Die area	100 mm²
Raw capacity	1.46 GB

Parameter	Value
ECC overhead (12.5%)	−0.183 GB
Bad-block reserve (2%)	−0.029 GB
User capacity	**1.236 GB** (displayed as 1.23 GB)

The user capacity of 1.236 GB is the design value used throughout this document.

Parameter	Value
Die cost (primary)	$6.85
User capacity	1.236 GB
Cost/GB primary	$5.54
Cost/GB volume	$3.34

XIII.4 Comparison with Existing Technologies

Technology	Cost/GB (user, 2026)
DDR5-8000	$5–8
DDR4	$5–10
NAND (SLC)	$0.50–1.00
NAND (TLC)	$0.15–0.30
MRAM (STT)	$3–5
Conventional FeRAM (PZT)	$2–5
Vortex FeRAM (primary)	**$5.54**
Vortex FeRAM (volume)	**$3.34**

The Vortex FeRAM sits at the DDR5 price level. It is **cheaper than DDR5 at primary pricing** ($5.54 vs $5–8, marginally), and **comparable at volume** ($3.34 vs $2.50–4.00). It is **5–11× more expensive than NAND SLC**. It is **comparable to MRAM** ($3–5).

XIII.5 Target Sell Price and Gross Margin

Segment	Die cost (primary)	Recommended TSP	Gross margin
Embedded / industrial	$6.85	$15.00	54%
Automotive	$6.85	$20.00	66%
Read-heavy storage	$6.85	$14.00	51%

These are die-level prices. Package, assembly, and test costs are included in the die cost above;

the TSP is the sell price to the OEM. Distribution and channel margins are not included in the table.

XIII.6 Cost-of-Ownership

XIII.6.1 Power cost

At $0.10/kWh and 100 mm² die:

State	Power	Annual energy	Annual electricity cost
Idle	109 mW	0.955 kWh/year	$0.10/year
50% duty (read-sat)	1021 mW	8.945 kWh/year	$0.89/year
100% duty (read-sat)	1934 mW	16.94 kWh/year	$1.69/year

For comparison, a DDR5-8000 16 GB DIMM at 4–6 W continuous draws 35–53 kWh/year and costs $3.50–$5.30/year in electricity. A 1.23 GB FeRAM die replaces less than one DDR5 DIMM in capacity, so the power comparison is not directly comparable at the die level. At the per-GB level, the Vortex FeRAM's idle power (66 mW/GB) is 4–6× lower than DDR5's idle power (250–375 mW/GB); at 50% duty, the Vortex FeRAM's power (621 mW/GB) is 1.7–2.5× higher than DDR5.

XIII.6.2 Total cost of ownership (5-year, embedded)

Item	Vortex FeRAM (1.23 GB)	DDR5-8000 (1 GB equivalent)
Device cost	$15	$5–8
Electricity (5 years, idle)	$0.50	$1.00–$1.50 (includes refresh)
Cooling (5 years, PUE 1.5)	$0.25	$0.50–$0.75
Refresh power (5 years)	$0 (non-volatile)	$1.00–$1.50
Total 5-year TCO	**$15.75**	**$7.50–$11.75**

The Vortex FeRAM has a higher device cost than DDR5 at the same capacity. Its TCO advantage is in applications where non-volatility and read latency are valued (the value is not captured in the device cost). For pure capacity-per-dollar, DDR5 is cheaper.

XIII.7 Volume Pricing

At volume above 100,000 units, wafer cost and package cost decrease:

Volume	Wafer cost reduction	Package cost reduction	Total die cost reduction
1,000–10,000 units	0%	0%	0%

Volume	Wafer cost reduction	Package cost reduction	Total die cost reduction
10,000–100,000 units	5%	10%	6%
100,000–1,000,000 units	10%	20%	12%
1,000,000+ units	20%	25%	22%

XIII.8 The Financial Case

The Vortex FeRAM is positioned for applications where non-volatility and read speed matter more than raw capacity-per-dollar. It is comparable to DDR5 on cost per GB ($5.54 vs $5–8 primary) and cheaper at volume ($3.34 vs $3.50–$4.00). It is not a NAND price competitor. The financial case rests on the value of non-volatility (no refresh power, instant power-off) and read latency (6–9× faster than DDR5), not on undercutting commodity DRAM pricing.

Part XIV — Advanced Optimization (Promotional Overview)

XIV.1 Overview

The Vortex FeRAM family includes a second product, the **Advanced product**, designed for applications where density and cost per GB dominate. The Advanced product is a higher-density variant of the Vortex FeRAM, optimized for read-heavy storage workloads. This Part provides a promotional overview only; the complete build and operation specification is published separately.

XIV.2 Advanced Product Summary

Parameter	Advanced product	vs Vortex FeRAM
Position	Density tier	Different segment
Effective density	31.25 GB/cm²	21×
User capacity (100 mm² die)	27.22 GB	22×
Read latency (on-chip, serial / pipelined)	4.78 ns / 2.70 ns	Comparable
Read latency (system)	~7.7–9.7 ns (serial)	Comparable
Read bandwidth	51.2 GB/s	Comparable
Write latency (avg / max)	13.5 / 15.5 ns	3.3× faster
Write latency (system)	~17–19 ns	~2.5× faster
Write bandwidth (pure-write)	2.37 GB/s	3.4×
Write energy per cell	0.267 fJ	3.3× lower
Endurance (1% fail, Tj = 87.2 °C)	5.14×10^5 cycles	4,800× lower
Endurance (0.1% fail)	1.62×10^5 cycles	4,800× lower
Retention at 85 °C	> 10 years	Comparable
Read-disturb margin (85 °C)	≥ 25×	≥ 25× guaranteed
Idle power	111 mW	+2 mW
50% duty power (read-sat)	1023 mW	+2 mW
100% duty power (read-sat)	1936 mW	+2 mW
100% duty thermal	Requires thermal pad (θ_{JA} = 15 °C/W) or ambient ≤80 °C	Same

Parameter	Advanced product	vs Vortex FeRAM
Operating temperature (writes)	−40 to 85 °C (no warm-up required)	Wider cold-start range
Package	BGA-324 (18×18, 0.8 mm pitch)	Same
I/O rate	6.4 Gbps, 64-bit	Same
Cost/GB primary	$0.76	7.3× cheaper
Cost/GB volume	$0.41	8× cheaper

XIV.3 When to Choose the Advanced Product

The Advanced product is recommended for applications where:

- **Density matters more than endurance.** 31.25 GB/cm² vs 1.46 GB/cm² for the Vortex FeRAM (21× higher density).
- **Cost per GB is a primary metric.** $0.76/GB primary vs $5.54/GB for the Vortex FeRAM (7.3× cheaper), and $0.41/GB volume vs $3.34/GB (8× cheaper).
- **Write latency matters.** 13.5 ns per word vs 45.2 ns for the Vortex FeRAM (3.3× faster).
- **Write throughput matters.** 2.37 GB/s pure-write vs 0.7 GB/s for the Vortex FeRAM (3.4× faster).
- **Read-heavy storage workloads.** Sequential read bandwidth is 51.2 GB/s (same as the Vortex FeRAM), but the higher density enables larger working sets per device.
- **Wear-leveled storage** is the deployment model. The Advanced product's cycle endurance is sufficient for wear-leveled consumer storage, but is not intended for hot-write applications.

The Advanced product is **not** recommended for applications where:

- Very high cycle endurance ($\geq 10^8$ cycles) is required. Use the Vortex FeRAM (2.45×10^9 cycles) or DRAM (10^{16} cycles).
- Extreme cold operation below −40 °C is required (neither product is specified below −40 °C storage).

XIV.4 When to Choose the Vortex FeRAM

The Vortex FeRAM is recommended for applications where:

- **Endurance is the dominant requirement.** 2.45×10^9 cycles vs the Advanced product's 5.14×10^5 cycles (4,800× higher).
- **Embedded, industrial, or automotive applications** where write-heavy workloads, hot-storage, or high-cycle operation is expected.
- **Simple deployment.** The Vortex FeRAM's endurance exceeds most workload lifetimes even at peak write rate, so no wear-leveling infrastructure is required at the

host level.

- **Comparable cost to DDR5** is acceptable, and non-volatility plus read speed justify the premium over NAND.

XIV.5 The Complete Family at a Glance

Metric	Vortex FeRAM	Advanced product
Position	Endurance tier	Density tier
Density	1.46 GB/cm²	31.25 GB/cm²
User capacity (100 mm²)	1.23 GB	27.22 GB
Read latency (system)	~7.7–9.7 ns	~7.7–9.7 ns
Write latency (system)	~48–50 ns	~17–19 ns
Read bandwidth	51.2 GB/s	51.2 GB/s
Write bandwidth	0.7 GB/s	2.37 GB/s
Endurance (1% fail)	2.45×10^9	5.14×10^5
Retention at 85 °C	>10 years	>10 years
Idle power	109 mW	111 mW
Cost/GB primary	$5.54	$0.76
Cost/GB volume	$3.34	$0.41

XIV.6 Product Selection Guidance

For applications that can use either product:

- **If the workload is write-heavy or requires ≥ 10^8 cycles:** Vortex FeRAM.
- **If the workload is write-light or read-heavy and density matters:** Advanced product.
- **If the workload is a mix:** deploy both. Use the Vortex FeRAM for hot data and the Advanced product for cold data. The two products can coexist in the same system on separate memory channels.

Licensing

Document License

This document is released under the **Creative Commons Attribution 4.0 International (CC BY 4.0)** license. Under CC BY 4.0:

- The document may be copied, distributed, and modified for any purpose, including commercial use.
- **Attribution is required.** Any user of the document must give appropriate credit to the author (Martin Hristov), provide a link to the license, and indicate if changes were made.
- Full license text: https://creativecommons.org/licenses/by/4.0/legalcode.

Patent-Free Release

This technology is released patent-free. No patents are claimed on any aspect of the design, the materials, the fabrication process, or the circuit. The technology is placed in the public domain for the benefit of all. The copyright license (CC BY 4.0) and the patent-free release are separate and independent: the copyright license governs the use of the document, the patent-free release governs the use of the technology.

Repository Access

Complete specifications, updates, and supporting materials are available at:

- **Mega:** https://mega.nz/folder/v35nVLAB#NbS00c8CL8ONTEHyMPe-Fg
- **Proton Drive:** https://drive.proton.me/urls/CMHBZEQKVG#ScNAG2APdoH2
- **Zenodo:** Zenodo search
- **GitHub:** https://github.com/MartinVHristov
- **OSF:** https://osf.io/user/9ps8x
- **Academia.edu:** https://independent.academia.edu/MartinHristov37
- **X:** https://x.com/MHristov37

Contact

For technical questions about this specification:

M.V.Hristov@proton.me

Closing Statement

The Product Summary

The Vortex FeRAM is a non-volatile memory designed for applications where non-volatility, read latency, and very high cycle endurance are the dominant requirements. Key capabilities:

- Density: 1.46 GB/cm² (75% array efficiency)
- User capacity: 1.23 GB per 100 mm² die
- Read latency: 4.71 ns serial / 2.63 ns pipelined on-chip; ~7.7–9.7 ns system (6–9× faster than DDR5, 5,500–7,100× faster than NAND)
- Write latency: 45.2 ns per word average, ~48–50 ns system (comparable to DDR5, ~2,000× faster than NAND)
- Read bandwidth: 51.2 GB/s
- Write bandwidth: 0.7 GB/s pure-write (write-verify-limited)
- Write energy: 0.88 fJ per cell (cell-level; system-level ~6.5 pJ per bit including I/O)
- Endurance: 2.45×10^9 cycles at 1% fail (24,500× NAND)
- Retention: >10 years at 85 °C
- Idle power: 109 mW (66 mW/GB)
- Cost per GB: $5.54 primary, $3.34 volume (comparable to DDR5)
- Non-volatile: >10 years retention, instant power-off, no refresh

What the Product Is

The Vortex FeRAM is a read-fast, non-volatile, endurance-focused memory. It is positioned for applications where non-volatility, read latency, and very high cycle endurance are the dominant requirements. It is fabricated on standard 28 nm CMOS with HZO ferroelectric atomic layer deposition.

What the Product Is Not

The Vortex FeRAM is not a DDR5 write-bandwidth replacement (write bandwidth 0.7 GB/s vs DDR5's 51 GB/s). It is not a DRAM endurance replacement (2.45×10^9 cycles vs DRAM's 10^{16}). It is not a NAND price competitor ($5.54/GB vs NAND's $0.50–1.00/GB). It is not a drop-in replacement for existing memory interfaces (custom protocol, custom controller required).

The Release Gates

The Vortex FeRAM ships contingent on four specific measurements:

1. **Core DC conductivity** — single-cell I-V measurement confirming that the

ferroelectric core carries DC current. If not, the read path is broken and the device requires 6–12 month redesign.

2. **Core radius** — single-cell measurement confirming the vortex core radius within 30% of the 2.317 nm design value.
3. **Dry etch-back anisotropy** — process qualification confirming the pillar dry etch achieves ≥ 3:1 anisotropy.
4. **CMP endpoint** — process qualification confirming the cap CMP stops at the HZO field with ≤ 1.0 nm Al_2O_3 overlap over the annular region.

All four gates are measurements or process qualifications, not design questions. The design is stable; the gates confirm the numbers.

The Complete Family

The Vortex FeRAM family includes a second product, the **Advanced product**, designed for density and throughput. The Advanced product is published as a separate engineering specification. The two products together cover the full spectrum from embedded/industrial (Vortex FeRAM) to high-volume storage (Advanced product).

The Release

This specification is released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. No patents are claimed on any aspect of the design, the materials, the fabrication process, or the circuit. The technology is placed in the public domain for the benefit of all. The engineering team can build and test the memory from this document alone. The software team can build the driver and the wear-leveling stack from Part XII. The applications team can select the appropriate product from the family comparison in Part XIV.

End of the Vortex FeRAM Engineering Specification.